Janus

Eleven modules for polyphonic voices, for Voltage Modular



Janus is a series of eleven modules that together make a polyphonic synthesiser: a key assigner, an oscillator, a ladder filter, an envelope generator, an LFO, a slew limiter, an output stage, a high-pass filter, a bucket-brigade chorus, and two CV converters.

Each voice can have its own pitch, modulation rate, timing, brightness and stereo position, and when several voices play one key, each of these can be spread across them by a set amount. This manual describes every control and jack of each module, and gives example patches. Its graphs are measured from the modules' own DSP code.

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Introduction

What the series is for

Janus is a series of eleven modules that together make a polyphonic synthesiser in Voltage Modular. Its sound is based on a family of polysynths from the early 1980s, which had a digitally timed oscillator, a resonant ladder filter and a bucket-brigade chorus. Those instruments are known for a muted timbre, a slow movement within held chords, and the faint hiss of the chorus.

The series is not a recreation of any of them. It uses their timbres and their familiar controls as a starting point for something new. Each module is named after the part of the classic voice it replaces. Like the god Janus, each module faces two ways: towards a sound and a way of patching that players already know, and towards what becomes possible when every part is a separate module.

Two rules apply to every module.

Every module loads as the classic part. With its default settings, each module behaves like the corresponding part of the classic voice. Every control the series adds has a setting at which it has no effect at all: at that setting the control is completely transparent, and the module loads with it there.

The additions only extend the classic part. A module can do more than the part it replaces, for example a polyphonic input where there was one control, an adjustable curve where there was a fixed shape, or extra filter responses. But each addition is something you can turn on as well as the classic behaviour, never something that replaces it.

Pulling out the control plane

In the instruments the series is based on, everything that decided how a note was played was built in: which voice a key went to, the envelope each voice followed, the single LFO sent to every voice, the portamento between notes. It ran in firmware on one circuit board. You could set its knobs, but you could not patch it.

Janus splits this control layer into separate modules, connected by cables. **KEY** assigns notes to voices, **ENV** makes an envelope for each voice, **LFO** makes an LFO for each voice, and **SLU** glides each voice from one value to the next. Wherever the classic voice had one value shared by all its voices, the Janus module has a polyphonic input, so each voice can have its own value.

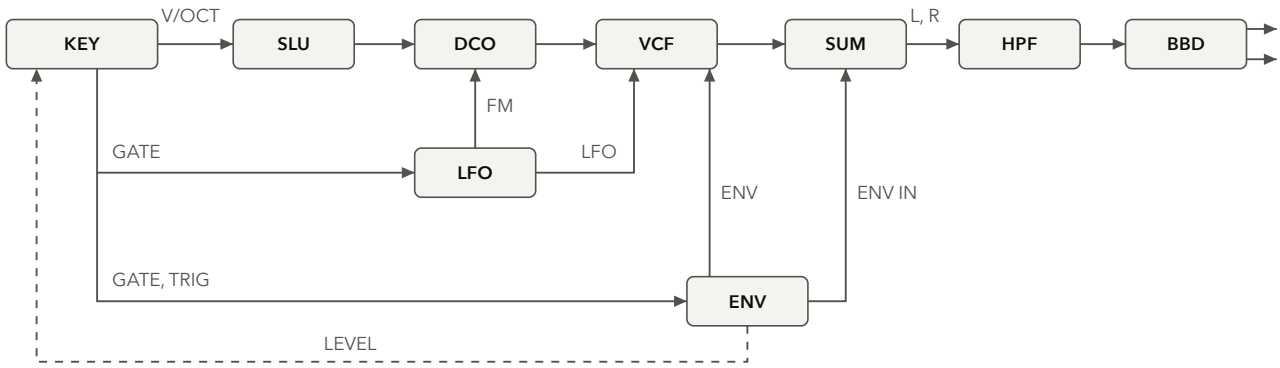


Figure 1. The classic voice, patched with the series. Every cable is polyphonic up to **SUM**, which mixes the voices to a stereo pair; **HPF** and **BBD** process the pair. The dashed cable sends the envelopes back to **KEY**, so that it can prefer released voices that have finished sounding. The per-voice VCA is inside **SUM**.

Stacks and fans

The usual way to make a supersaw is to play several sawtooths on one note, each detuned a little from the others. It sounds thick because the voices are never exactly in tune: they drift in and out of phase, and the beating between them keeps changing.

Janus applies the same idea to more than pitch. **KEY** can give each key a *stack* of voices instead of one: from 1 to 16 lanes of the poly cable. More voices per key means fewer keys can play at once: sixteen keys of one voice, four keys of four, or one key of sixteen. **DETUNE** spreads the stack's pitches, as in a supersaw. KEY also outputs each voice's position in its stack as a voltage, on the **FAN** outputs: −5 V at one end of the stack and +5 V at the other, spaced by one of three laws (Figure 2).

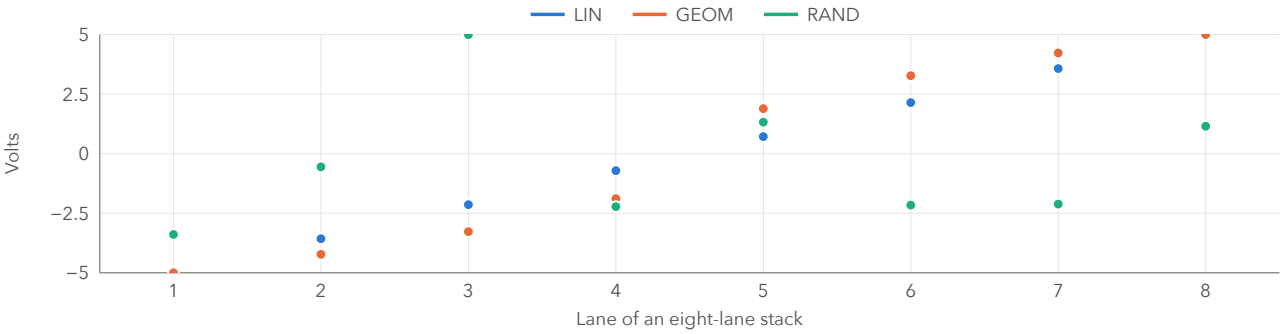


Figure 2. KEY's three FAN outputs for one key held at WIDTH 8, measured. LIN spaces the lanes evenly. GEOM pushes them towards the ends, leaving a gap in the middle, the shape of a supersaw. RAND gives each lane a random position when the key is played, and keeps it for the note.

A fan output is a modulation source that gives each lane's position within its stack, which no other module can work out for itself. Patch it into any polyphonic input, and that control is spread across the stack in the same way DETUNE spreads the pitch. For example:

- into LFO's **RATE POLY** input, each voice's vibrato runs at a slightly different rate, so the vibratos drift apart and back together (Figure 3);
- into LFO's **DELAY POLY** input, each voice's vibrato fades in over a different time;
- into SLU's time inputs, each voice glides at a different speed;
- into the oscillator's pulse width, the filter's cutoff or SUM's stereo position, each voice gets a different width, brightness or place.

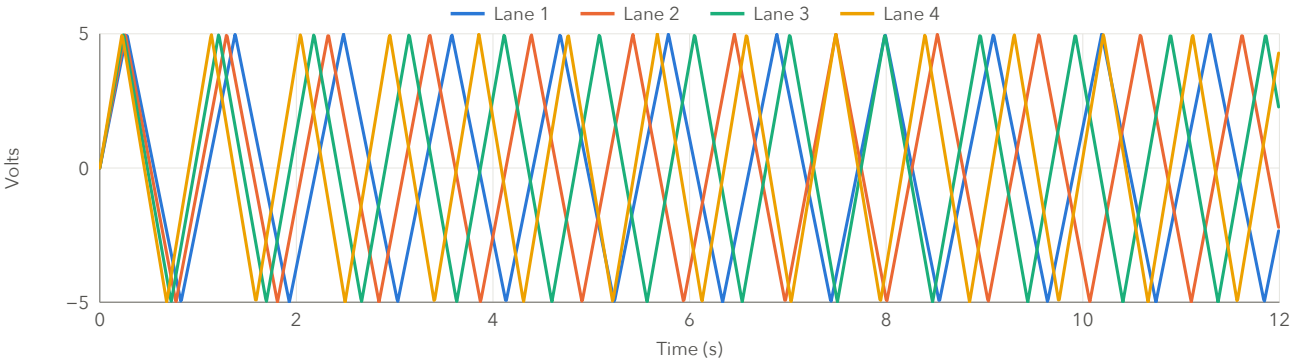


Figure 3. A four-lane stack with KEY's LIN fan patched into LFO's RATE POLY input. All four lanes start together on the note, and run at 0.91, 0.97, 1.03 and 1.10 Hz. The outer two are half a cycle apart after 2.6 s, and back in step after 5.1 s.

Analogue polysynths sound lively partly because their voices were never quite identical: slightly different tuning, slightly different envelope and LFO timing. In hardware this variation came from component tolerances, and you could not control it. In Janus you set it: how much variation, on which controls, with which spacing. It is the same every time the patch is loaded, and it can itself be modulated.

Three more modules work with fans. **MTP** turns any mono signal into a fan, with the same three spacing laws, so that a single LFO or knob can be spread across the voices. **CVC** reshapes a fan. For example, it can fold the fan at 0 V so that the two outer voices of a stack get the same value and the inner voices get another, which usually suits a filter cutoff better than an unfolded fan. **SUM** can use a fan as the stereo position of each voice.

The eleven

Module	What it is
KEY	MIDI to sixteen voices, with stacks, detune, and the fans
MTP, **CVC**	Two converters: a mono signal spread across a poly cable, and a poly signal folded, curved and offset
DCO	The oscillator: saw, pulse, sub-octave square and noise, mixed on the panel, with phase modulation, through-zero FM and sync
VCF	A four-pole ladder filter per voice, with drive, four responses and two slopes
ENV	An ADSR per voice, with a retrigger input, two curves and a stepped scan
LFO	An LFO per voice, with a fade-in that each note or each phrase can start
SLU	A slew limiter per voice, for glides and for shaping any control
SUM	The output stage: per-voice VCAs, a tremolo stage, a stereo field across the voices, meters
HPF	A four-position high-pass for the summed pair
BBD	A stereo bucket-brigade chorus, with its hiss, drift and ripple

The chapters start with KEY and the two converters, then DCO and VCF, the modulation modules ENV, LFO and SLU, and the output stage. Chapter 11 gives example patches, and the two appendices describe how the chorus and the oscillator work.

What the eleven share

Sixteen lanes. Every polyphonic module handles up to sixteen voices, one per lane of a poly cable, and processes as many lanes as the patch's polyphony setting. Each lane is processed separately.

Pitch. 1 V per octave, and 0 V is C2 (65.41 Hz), as Voltage Modular has it.

Levels. Audio and CV are both ±5 V at full scale. A gate is high when it is above 0 V; KEY's gates are 5 V. DCO's four sources added together can exceed ±5 V. The modules do not limit their outputs unless the chapter says so.

Mono and poly inputs in pairs. Voltage Modular does not connect a mono cable to a poly jack, so most modulation inputs come in pairs: **MONO** takes one signal applied to every voice, and **POLY** takes a separate signal for each voice. The two are added together, and on most modules one **AMT** knob scales the sum. AMT has no effect at its centre, where it loads; turned right it applies the CV, and turned left it applies the CV inverted. The chapters describe the exceptions. KEY's **COMMON** outputs are mono, for patching to mono inputs. MTP converts a mono signal to a poly one.

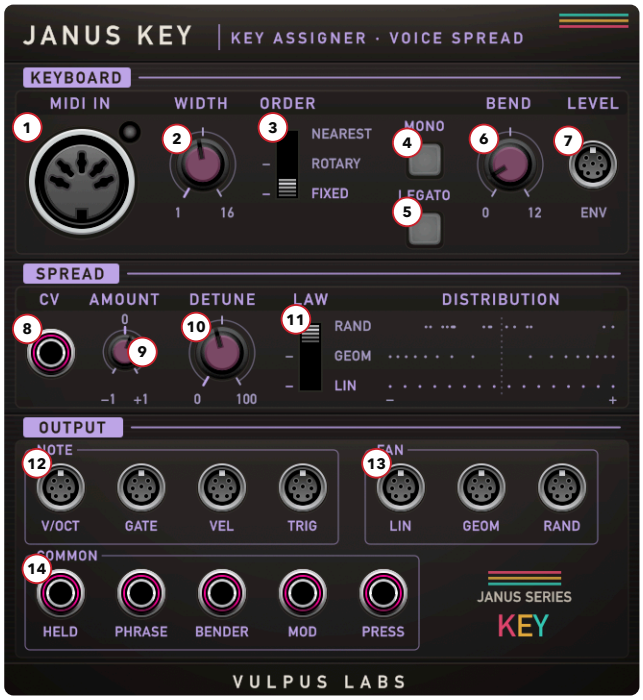
Default settings. Each module loads with the settings of the classic voice, and each added control at the setting where it has no effect. The chapters give the defaults.

Tooltips. Hovering over most controls shows the value in its own units: hertz, cents, decibels, seconds or volts. On many controls you can click the tooltip and type a value, in the same units: `250` on a time sets 250 ms, and `440` on a cutoff sets 440 Hz. A value outside the control's range is set to the nearest end of the range.

1 · KEY

KEY ASSIGNER · VOICE SPREAD

KEY converts one MIDI input into up to sixteen voices. With only **MIDI IN**, **V/OCT** and **GATE** patched, it is an ordinary MIDI-to-CV converter. Its other controls set which voice each key is given, and how the voices given to one key differ from each other.



1	**MIDI IN**	notes, bend, mod wheel, pedal, pressure
2	**WIDTH**	voices per key, 1 to 16
3	**ORDER**	which free voices a key is given
4	**MONO**	one key at a time
5	**LEGATO**	overlapping keys do not retrigger
6	**BEND**	pitch bend range, 0 to 12 semitones
7	**LEVEL**	envelope levels, fed back
8	**CV**	modulates DETUNE
9	**AMOUNT**	the CV's amount, ±
10	**DETUNE**	pitch spread of a stack, 0 to 100 cents
11	**LAW**	LIN, GEOM or RAND
12	**NOTE**	V/OCT, GATE, VEL, TRIG, per voice
13	**FAN**	LIN, GEOM, RAND, per voice
14	**COMMON**	HELD, PHRASE, BENDER, MOD, PRESS

MIDI IN

MIDI IN reads note on and off, pitch bend, the mod wheel (CC1), the sustain pedal (CC64), channel pressure, and the two panic messages, all notes off and all sound off (CC123 and CC120). It accepts every MIDI channel. The lamp beside it lights when a message arrives.

A transport start, continue, stop or reset message releases every held note. A DAW often sends no note-offs when it is stopped mid-phrase, and without this the notes would stay on.

Stacks

Every control loads at its lowest setting: WIDTH 1, ORDER at FIXED, MONO and LEGATO off, DETUNE at 0, AMOUNT at its centre, and LAW at LIN. BEND loads at 1. The number of lanes is the patch's polyphony setting, up to sixteen; the examples in this chapter assume sixteen.

WIDTH sets how many adjacent lanes, from 1 to 16, are given to each key. We call these lanes a *stack*. KEY always handles a stack as a unit: a note-off releases the whole stack, voice stealing takes a whole stack, and a note never keeps only part of its stack.

At 1, KEY is an ordinary assigner, one voice per key. At 16, all sixteen lanes play one key. In between, more voices per key means fewer keys can sound at once. The tooltip shows the result rather than the number, for example `4 stacks of 4 voices`. If the width does not divide sixteen exactly, the leftover lanes are not used, and the tooltip says so: a width of 6 shows `2 stacks of 6 voices, 4 dark`.

A change of WIDTH takes effect from the next note. Changing the layout of notes already held would cause a click on every lane.

Which stack a key is given

ORDER sets which free stack a new key is given. It makes the most difference when you play faster than notes release.

- **FIXED** gives the lowest-numbered free stack. A repeated key gets the same stack each time, which cuts off its own release. This is the most predictable order.
- **ROTARY** gives the next stack after the last one used. A repeated key moves through the stacks, so each note's release keeps sounding while the next note plays.
- **NEAREST** gives the stack whose last note was closest in pitch; stacks that have not played yet are used first, lowest first. Use it when a slew limiter is patched after V/OCT: each lane glides from the last pitch it played, so NEAREST gives each new key the shortest glide.

A voice is free as soon as its key is released, even if its release stage is still sounding. To avoid cutting off releases, patch **LEVEL**.

Voice stealing and the LEVEL input

LEVEL takes the envelope outputs back from ENV, as a poly cable. It changes which voice is stolen when a new note needs one, not how anything sounds. With LEVEL patched, KEY chooses as follows, in this order:

1. A repeated key takes back its own voice first, if it has been released (not under ROTARY).
2. A voice that has finished is taken before one that is still sounding. A voice counts as finished once its level has been under 5 mV for a continuous millisecond. ORDER chooses among finished voices.
3. If every released voice is still sounding, the quietest is taken.

With or without LEVEL, if no voice has been released, the oldest note is taken; a note held only by the sustain pedal is taken before one whose key is down. Under NEAREST, closeness in pitch takes priority over rules 2 and 3. When a note needs a stack and none is free, KEY releases whole stacks; it never takes some of a held note's lanes.

MONO and LEGATO

MONO plays one key at a time, on one stack of WIDTH lanes, and leaves the other lanes silent. MONO with WIDTH at 1 is a single voice; with WIDTH at 8 it is eight voices in unison. The most recent key has priority. When you release it, the most recent key that is still held plays again.

LEGATO only has an effect in MONO. When it is off, every key starts a new note: a new TRIG, a new RAND draw, and a restarted envelope. When it is on, a key played while another is still held only changes the pitch and VEL, and the note continues. In both cases the gate stays high while any key is held. The first note after all keys have been released always starts a new note.

BEND sets the pitch bend range, 0 to 12 semitones in each direction, in whole semitones. It loads at 1. The **BENDER** output gives the wheel's position, and is not affected by BEND.

Spread

The **SPREAD** section sets how the lanes of a stack differ in pitch. At a width of 1 it has no effect, except with LAW at RAND.

DETUNE sets the total pitch spread of a stack, 0 to 100 cents. At 100 cents the outer lanes are 50 cents either side of the note. A few cents makes the sound thicker; twenty cents is clearly detuned. Changes to DETUNE apply to notes already held.

CV and **AMOUNT** modulate DETUNE. AMOUNT is an attenuverter: it has no effect at its centre, turned right it applies the CV, and turned left it applies the CV inverted, so a rising envelope can narrow the stack instead of widening it. With AMOUNT fully right, +5 V sets the full 100 cents and −5 V sets no spread, wherever the knob is; fully left, the reverse; the result is limited to 0 and 100 cents. The input is mono, because it sets the width of the whole stack. To vary something per lane, use the LAW and FAN outputs.

LAW sets how the lanes are spaced within the spread. The **DISTRIBUTION** display beside it shows each law as sixteen dots, from − to +.

- **LIN** spaces the lanes evenly.
- **GEOM** moves the lanes towards the ends, so they are close together at the ends and far apart in the middle. This is the spacing of a supersaw.
- **RAND** gives each lane a random pitch offset when the note starts, and keeps it until the note ends. Unlike the other two laws, RAND also detunes a single voice at a width of 1, so each note plays slightly out of tune, by a different amount each time.

A change of LAW takes effect from the next note.

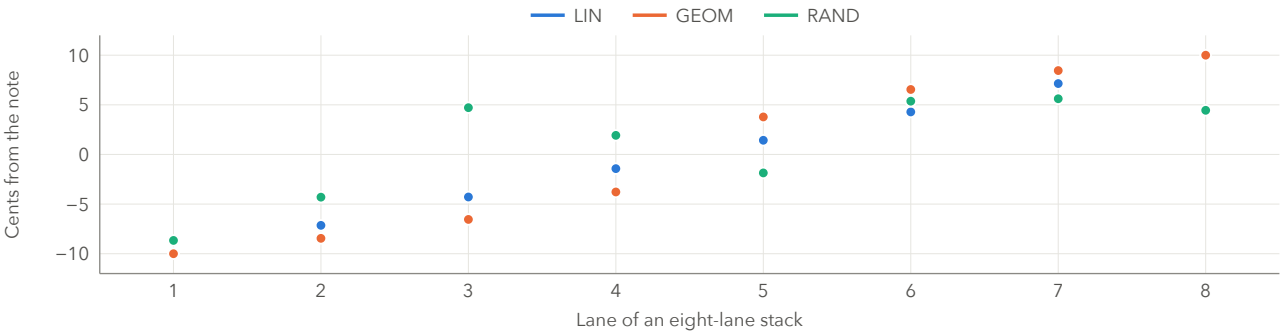


Figure 4. Each lane's pitch at V/OCT with DETUNE at 20 cents, measured. The outer lanes are 10 cents either side of the note. LIN steps by 2.86 cents per lane; GEOM leaves a gap of 7.6 cents in the middle. RAND's pitch offsets are drawn separately from the RAND fan's positions.

Outputs

NOTE, one per voice

Jack	What it carries
V/OCT	pitch, 1 V/oct, including detune and bend
GATE	5 V while the key or the sustain pedal holds the note
VEL	velocity, 0 to 5 V, held for the length of the note
TRIG	a one-sample 5 V pulse at the start of every note

TRIG also fires when a voice is given a new note while its gate is still high, which the gate alone cannot show. Patch GATE to an envelope's gate input and TRIG to its retrigger input.

When a key is released, its voice's gate goes low, but its pitch and fan outputs keep their values, because the note is still sounding through its release. If the pitch dropped to 0 V on release, the release would sweep down in pitch.

FAN, one per voice

LIN, **GEOM** and **RAND** give each lane's position in its stack, from −5 V at one end to +5 V at the other, spaced by that law (Figure 2). All three are always active, and changing LAW does not change them. The values are set when the key is played and held until the note ends.

There are no amount controls on the fan outputs: the inputs they are patched into have their own. The introduction and chapter 11 show some uses.

COMMON, one per keyboard

Jack	What it carries
HELD	5 V while any voice is held
PHRASE	a one-sample 5 V pulse when the first key is played after all keys were released
BENDER	the pitch wheel's position, ±5 V, exactly 0 V at centre
MOD	the mod wheel, 0 to 5 V

PRESS	channel pressure, 0 to 5 V

Use these outputs for mono inputs. If you patch a poly output into a mono input, the input reads lane 1 only, which goes silent whenever a note is played on another lane. PHRASE does not fire when a note is added to keys already held; for a pulse on every note, use TRIG.

Starting points

MIDI-to-CV. Patch MIDI IN, V/OCT, GATE and VEL, and leave every control at its default. One voice per key, no spread.

Unison lead. MONO on, WIDTH 8, DETUNE about 20 cents, LAW GEOM, with eight lanes into DCO. Turn LEGATO on so that overlapping notes change pitch without restarting the envelope. For a glide, patch SLU after V/OCT.

Thicker chords. WIDTH 4, DETUNE about 12 cents, LAW LIN: four keys of four voices each. To vary the brightness across each stack, patch the LIN FAN output through CVC into the filter (chapter 11).

Polyphonic glide. ORDER NEAREST, with SLU between V/OCT and the oscillator. Each key goes to the voice with the shortest glide.

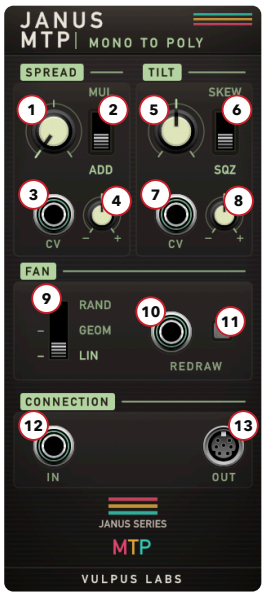
A random value per note. WIDTH 1, with the RAND FAN output into a control that should vary from note to note, such as stereo position, pulse width or filter cutoff. A few cents of DETUNE with LAW at RAND also gives each note a small random tuning error.

2 · MTP and CVC

MONO TO POLY · CV CONVERTER

Two small utility modules. **MTP** converts a mono signal to a poly one, giving each lane a value within a range around the input. **CVC** folds, curves, scales and offsets every lane of a poly signal. Neither replaces a part of the classic voice: they are needed because the voice is split into separate modules connected by cables.

MTP



1	**SPREAD**	width of the range, 0 to 1
2	**ADD, MUL**	range in volts, or relative to the input
3	**CV**	modulates SPREAD
4	**AMT**	the CV's amount
5	**TILT**	moves the lanes within the range
6	**SQZ, SKEW**	how TILT moves them
7	**CV**	modulates TILT
8	**AMT**	the CV's amount
9	**FAN**	LIN, GEOM or RAND
10	**REDRAW**	new random values, on a trigger
11	**Button**	new random values, by hand
12	**IN**	mono
13	**OUT**	poly, one lane per voice

A poly patch often needs one value for every voice, such as a filter cutoff, a pitch offset or an LFO. If every voice gets exactly the same value, sixteen voices can sound like one loud voice. MTP gives each voice a slightly different value, by an amount you set. For example: a few millivolts of spread on a pitch detunes a stack of oscillators; a volt or two on a cutoff gives each voice a different brightness; an LFO through MTP moves every voice together, each from its own offset.

With SPREAD at 0, every lane equals the input, so MTP has no effect until you turn SPREAD up. The output has one lane for each voice of the patch's polyphony. If the polyphony changes, the lanes are spread again across the new number of voices.

SPREAD

SPREAD sets the width of the range, from 0 to 1. The knob is not linear: its halfway point is 0.1, so the first half of its travel gives fine control of small spreads. The tooltip shows three decimal places.

The **ADD** and **MUL** switch sets what SPREAD is measured against.

- **ADD** (lower position): SPREAD is a fraction of 5 V. At 1, the lanes range from 5 V below the input to 5 V above it, limited to ±5 V (see Limits).
- **MUL**: SPREAD is a fraction of the input voltage. At 1, the lanes range from 0 V to twice the input. The range shrinks to zero as the input approaches 0 V. With a negative input, the lanes are in the same order as with a positive one: the first lane is always the lowest.

CV and **AMT** modulate SPREAD. At full AMT, 5 V covers the knob's whole travel. The result is limited to 0 and 1.

SPREAD, ADD	Each side of the input	On a pitch CV
0.002	10 mV	12 cents
0.010	50 mV	60 cents

0.100	0.5 V	half an octave
1.000	5 V	five octaves

TILT

TILT moves the lanes within the range, from −1 to +1. At its centre, where it loads, the lanes are spread evenly across the whole range. The **SQZ** and **SKEW** switch sets how TILT moves them.

- **SQZ** (lower position) shrinks the range towards one end. At +0.5, the lanes fill the top half of the range, from the input upwards; at +1, every lane is at the top. The lanes stay evenly spaced. Negative values do the same towards the bottom.
- **SKEW** keeps the top and bottom lanes where they are and moves the others towards one end. At +1, the middle lane is 90% of the way to the top.

Use SQZ to put every voice on one side of the input, for example every voice brighter than the input and none darker. Use SKEW to keep the full range but change how many voices are near each end, for example one dark voice and the rest bright. **CV** and **AMT** work as they do for SPREAD: at full AMT, a CV from −5 to +5 V sweeps TILT across its whole range.

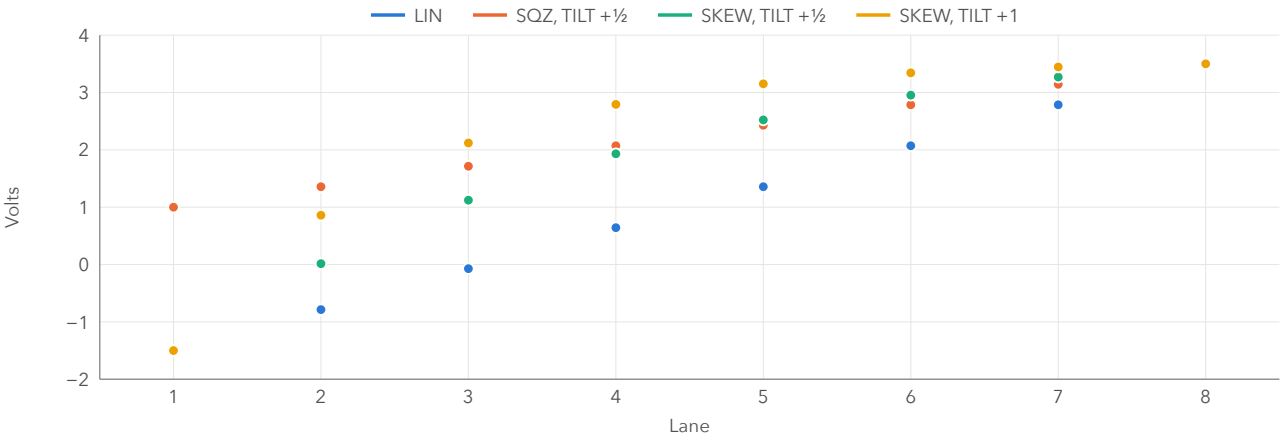


Figure 5. Eight lanes, 1 V in, SPREAD 0.5 on ADD, so the range is −1.5 to +3.5 V, measured. SQZ at +0.5 puts the lanes between the input and the top. SKEW keeps the end lanes fixed; at +1, lanes 3 to 8 are all above +2 V.

FAN and REDRAW

FAN sets how the lanes are spaced before TILT is applied. LIN and GEOM space the lanes as KEY's laws of the same names do; RAND differs, in that KEY draws a new position at each note and MTP keeps its positions until a redraw.

- **LIN** (bottom): evenly, from the bottom of the range to the top.
- **GEOM**: the end lanes stay at the ends and the others move outwards, so the lanes are close together near the ends and far apart in the middle.
- **RAND**: each lane gets a random position in the range, which it keeps until the next redraw. The positions are not sorted, so the first lane can be higher than the last. A lane keeps its position if the polyphony changes: going from eight voices to four leaves the first four unchanged.

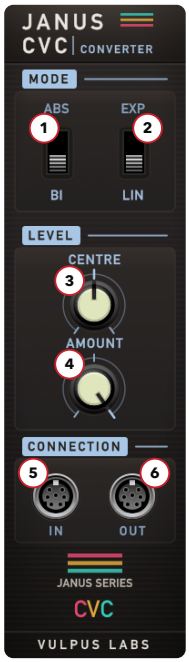
A new MTP always starts with the same random positions, and they are saved with the patch, so a patch sounds the same each time it is loaded. **REDRAW** chooses new random positions when its input rises above 0 V, or when you press its button. Under LIN and GEOM a redraw has no audible effect, but the new positions are kept and used when FAN is switched to RAND.

Limits

The input is limited to ±5 V before anything else, and the output never goes beyond ±5 V: a lane that would exceed it is held at ±5 V. With one voice under LIN or GEOM, the single lane is placed in the middle of the range, which with TILT at its centre equals the input; under RAND it keeps its random position. Lanes above the patch's polyphony are

set to 0 V. The knobs move to a new setting over a few milliseconds, so turning one quickly does not step every lane at once; the CVs and switches take effect immediately. An input that is not a valid number is read as 0 V.

CVC



1. **BI, ABS** — whether the input is folded at 0 V
2. **LIN, EXP** — linear or exponential curve
3. **CENTRE** — output for 0 V in, ±5 V
4. **AMOUNT** — gain, −1 to +1
5. **IN** — poly
6. **OUT** — poly

KEY's fan outputs range from −5 V to +5 V across a stack. That suits a stereo position, but not a filter cutoff: patched directly, a fan brightens the voices at one end of the stack and darkens those at the other. Usually you want the two outer voices to get the same value, and the two inner voices another. The filter cannot do that by itself, so CVC converts the signal before it reaches the filter.

BI passes the input with its sign. CVC is then a curve, an offset and an attenuverter for a poly signal. **ABS** takes the absolute value of the input: −5 V and +5 V both become +5 V, and 0 V stays 0 V. Applied to a fan, ABS gives each voice's distance from the middle of its stack.

LIN is a straight line. **EXP** is a curve that rises slowly at first and quickly at the end, the same curve family as ENV's curves and SUM's velocity curve. A half-scale input gives only a few per cent of full scale out. LIN and EXP give the same output at 0 V and at full scale. Under BI the curve is applied symmetrically either side of 0 V, so a bipolar signal stays bipolar.

AMOUNT multiplies the converted signal by −1 to +1. It loads fully clockwise (×1), not at its centre like the other modules' attenuverters, because here the converted signal is the module's whole output: at the centre the module would output only CENTRE. Below the centre it inverts the signal; under ABS the result is still folded.

CENTRE sets the output voltage for a 0 V input, from −5 V to +5 V. With nothing patched, CVC outputs CENTRE as a constant. With AMOUNT at its centre, the output is CENTRE.

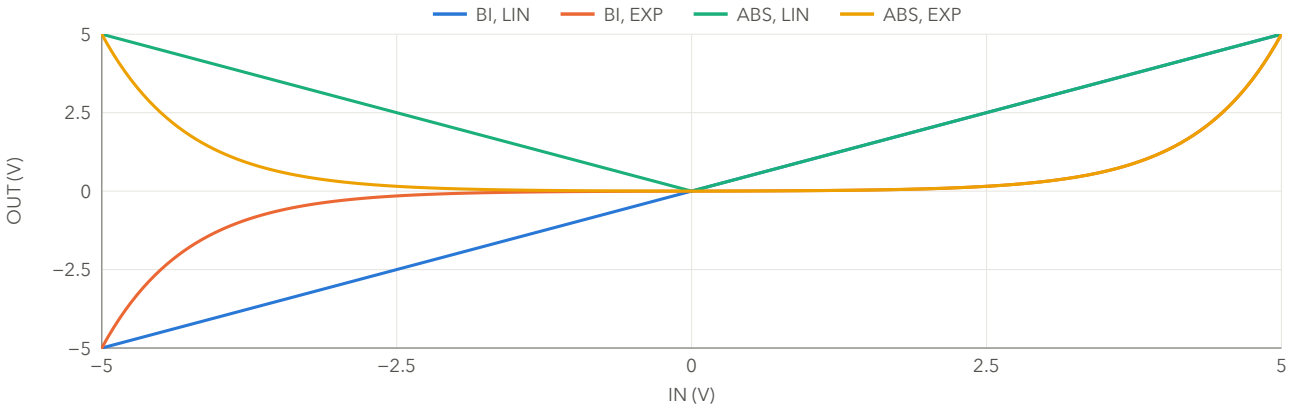


Figure 6. CVC's four transfer curves, AMOUNT fully clockwise and CENTRE at 0 V, measured. EXP is steep: +2.5 V in gives +0.15 V out, and the output reaches 1 V only at about 3.85 V in.

With BI, LIN, CENTRE at 0 V and AMOUNT fully clockwise, CVC is completely transparent: the output is identical to the input, within ±5 V. You can add it to a working patch, and it changes nothing until you change a setting.

Example: a fan into a filter cutoff

Four voices, with KEY's LIN fan output into IN, and OUT into the filter's cutoff.

	Lane 1	Lane 2	Lane 3	Lane 4
FAN	−5.00	−1.67	+1.67	+5.00
BI	−5.00	−1.67	+1.67	+5.00
ABS, AMOUNT +1, CENTRE 0 V	+5.00	+1.67	+1.67	+5.00
ABS, AMOUNT +1, CENTRE −5 V	0.00	−3.33	−3.33	0.00
ABS, AMOUNT −1, CENTRE +5 V	0.00	+3.33	+3.33	0.00

With BI, two voices are brightened and two darkened. With ABS, the two outer voices are brightened equally and the two inner voices less. The last two rows leave the outer voices unchanged and darken or brighten the inner ones.

Limits

The output is limited to ±5 V, whatever the settings. The limit is applied after CENTRE is added, so CENTRE can push the output to ±5 V. An input that is not a valid number gives CENTRE. Lanes above the patch's polyphony are set to 0 V.

3 · DCO

DIGITALLY CONTROLLED OSCILLATOR

DCO is sixteen oscillators on one poly cable. Each voice makes a sawtooth, a pulse of variable width, a square one octave below, and noise. Four faders mix them to **MIX OUT**. The pitch is timed digitally, so it does not drift. Pulse width, FM and phase modulation each have a per-voice CV input, and **SYNC** resets each voice from its own input.



1	**V/OCT**	pitch, per voice
2	**OCTAVE**	−4 to +4
3	**SEMITONE**	−12 to +12
4	**CENTS**	−100 to +100
5	**ANTI-ALIAS**	runs at twice the rate
6	**WIDTH**	the pulse's width
7	**AMT**	pulse width CV amount
8	**POLY CV**	pulse width, per voice
9	**CV**	pulse width, all voices
10	**AMT**	FM amount
11	**LIN, EXP**	FM in hertz or in octaves
12	**POLY CV, CV**	FM, per voice and all voices
13	**AMT**	phase modulation amount
14	**POLY CV**	phase modulation, per voice
15	**SYNC**	restarts each voice
16	**Faders**	saw, pulse, sub, noise
17	**CV**	modulates each fader
18	**MIX OUT**	the four, mixed
19	**LEVEL**	output level, 0 to −24 dB

TUNE

V/OCT takes a pitch for each voice, at 1 V/oct, with 0 V at C2. Tracking is exact, and there is no trim. **OCTAVE**, **SEMITONE** and **CENTS** are added to it. OCTAVE and SEMITONE move in whole steps; CENTS moves in whole cents, for detuning one DCO against another.

The four sources

Source	What it is
Saw	bright, with all harmonics; for strings, brass and basses
Pulse	hollow as a square, thinner and brighter as it narrows; set by PULSE WIDTH
Sub	a square one octave below the saw, in phase with it; adds low end
Noise	white noise, for breath on pads and for percussion

The saw, pulse and sub all come from one ramp per voice. The saw is the ramp. The pulse is high while the ramp is above a threshold set by PULSE WIDTH, so it goes high part-way through the cycle and goes low when the saw resets. The sub changes state each time the saw resets, which makes it an exact square one octave down, always in phase with the other two (Figure 7). Appendix B describes how this works.

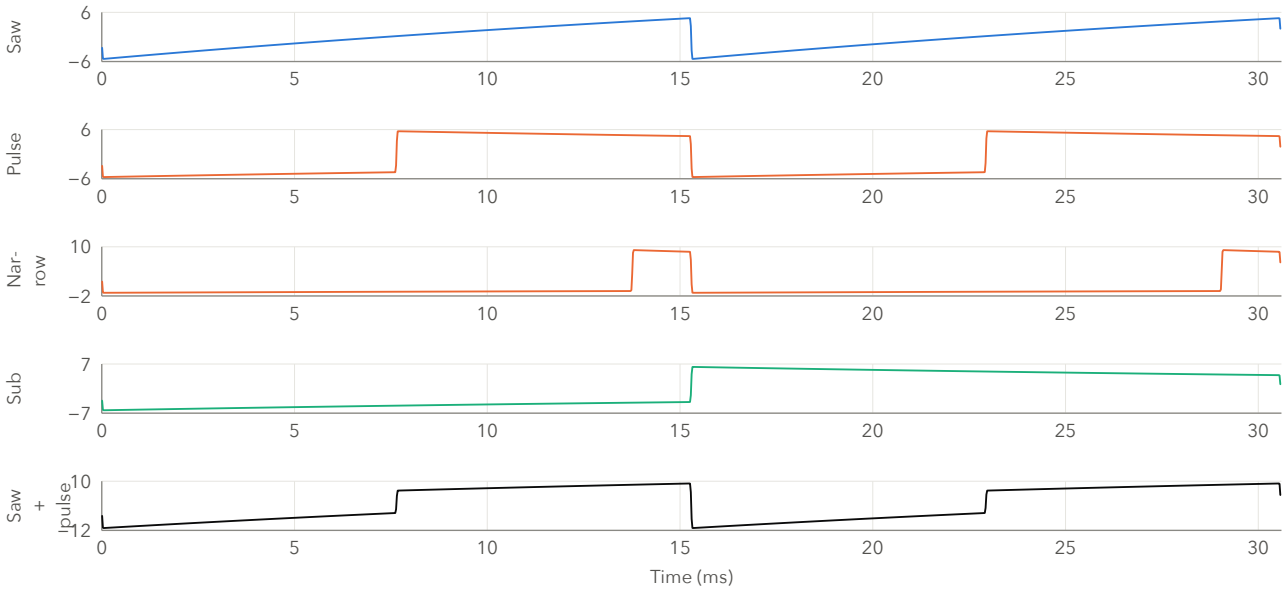


Figure 7. Two cycles of each source at C2, measured at MIX OUT with that source's fader at 10 (both faders for saw + pulse) and the rest at 0. With WIDTH at its centre the pulse is a square; the narrow pulse is WIDTH at 0.1, high for a tenth of each cycle. Saw and pulse together make a stepped saw: the pulse's rising edge adds a second step part-way up the ramp, and WIDTH moves that step. At this low pitch the output's 5 Hz high-pass filter (see Notes on the sound) visibly curves the saw and the sub.

The mix. Each fader runs from 0 to 10. The **CV** input above each fader is added to the fader's position, and a full-scale CV covers the fader's whole range, so with the fader at 0 an envelope can fade its source in. Each source peaks at about ±5 V on its own. The saw, pulse and sub can all be at their extremes at once, at the saw's reset, so all four faders at 10 can give about ±20 V. The output is not limited.

LEVEL, beside MIX OUT, sets the output level from 0 dB (fully clockwise, where it loads) down to −24 dB, without changing the balance between the faders. Use it to keep the level into VCF consistent. Sixteen voices on the same note are up to 24 dB louder than one voice if they are in phase (for example under SYNC), and sixteen detuned voices about 12 dB louder, which drives a saturating filter such as VCF much harder. So the knob's range covers one voice to sixteen voices in phase, and its centre suits sixteen detuned voices. At 0 dB it has no effect.

All voices share one noise source. Playing more notes makes the noise louder: sixteen voices of noise are sixteen times the level of one, rather than four times as with independent sources. Set the noise fader for the number of notes you will play.

MODULATION

Each section has a **POLY CV** input, which takes a value for each voice; the PULSE WIDTH and FREQ MOD sections also have a mono **CV** input. The two are added together and scaled by **AMT**, which has no effect at its centre and inverts the CV when turned left.

PULSE WIDTH

WIDTH sets the pulse width. At its centre the pulse is a square, which sounds hollow. Turning it either way makes the pulse narrower, which sounds thinner and brighter. The range is 5% to 95%; 5% and 95% sound the same, and the symmetrical range lets modulation move the width either way from the centre. For the shimmer of a classic string pad, patch a slow LFO into CV with AMT about a third of the way up. A narrow pulse is quieter than a square, so a width sweep also changes the level. WIDTH does not affect the sub.

FREQ MOD

FM MODE selects one of two kinds of frequency modulation.

- **EXP** (down): modulation is in octaves. Full-scale CV at full AMT moves the pitch one octave either way. Use it for vibrato and pitch envelopes. AMT has finer control near its centre, so small vibrato depths are easy to set.

- **LIN** (up): linear, through-zero FM. The modulation is added to the frequency in hertz, scaled to the note's frequency so that the effect sounds the same on every note. At full depth the frequency goes below zero and the waveform plays backwards. It gives harsher, less predictable sounds than EXP.

FREQ MOD is meant for LFOs, envelopes and pitch bend. For audio-rate modulation, use PHASE MOD.

PHASE MOD and SYNC

AMT and **POLY CV** shift each voice's phase, by up to one cycle either way. With another oscillator patched in, this gives FM-style tones: bells, mallets, metallic and glassy sounds. It stays clean when the modulator is at audio rate. A second DCO tuned an octave or a fifth higher is a good modulator, with an envelope controlling its level.

SYNC restarts each voice's waveform whenever that voice's SYNC input rises through 0 V. Patch a second oscillator into SYNC and sweep this DCO's tuning for the classic sync lead sound (Figure 8). The input is polyphonic, so each voice can be synced to a different source.

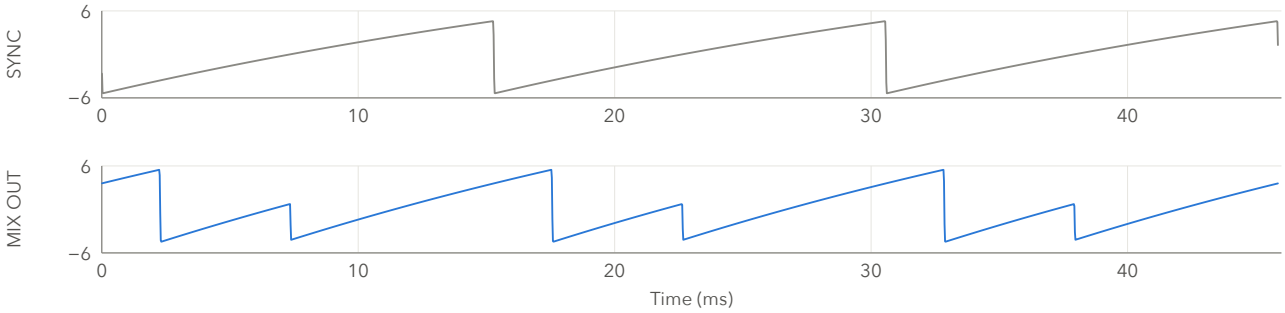


Figure 8. A saw tuned seven semitones up, synced to a C2 saw. Each time the SYNC input rises through 0 V, the voice restarts its cycle on the next sample. Each 15.3 ms cycle of the C2 saw contains one full 10.2 ms cycle of the synced voice and half of another.

ANTI-ALIAS

ANTI-ALIAS, the small button at the right end of the TUNE section, runs every voice at twice the sample rate, then filters out everything above about 21 kHz before converting back to the host's rate. This removes most aliasing. It is off by default, and when off, DCO is unchanged.

Turn it on together with VCF's ANTI-ALIAS button. Most of DCO's own aliasing is between 15 and 24 kHz, where it is inaudible, but a saturating filter such as VCF mixes it down into the audible range. With both buttons on, the series' measurements find almost no aliasing in a whole voice.

Inharmonic energy below 15 kHz, saw at A4	Off	On
DCO alone	−67.9 dB	**−90.1 dB**
Through VCF with its ANTI-ALIAS on, filter open	−74.4 dB	**−90.5 dB**
The same, VCF's DRIVE at +12 dB	−72.5 dB	**−88.8 dB**
Through VCF with its ANTI-ALIAS off, filter open	−39.5 dB	−41.3 dB

The last row shows why: a filter running at the host's sample rate produces much more aliasing itself than DCO sends it, so turning on DCO's ANTI-ALIAS alone makes little difference through it. Patched directly to an output, or into a module that does not saturate, DCO's ANTI-ALIAS improves aliasing by the full 22 dB on its own.

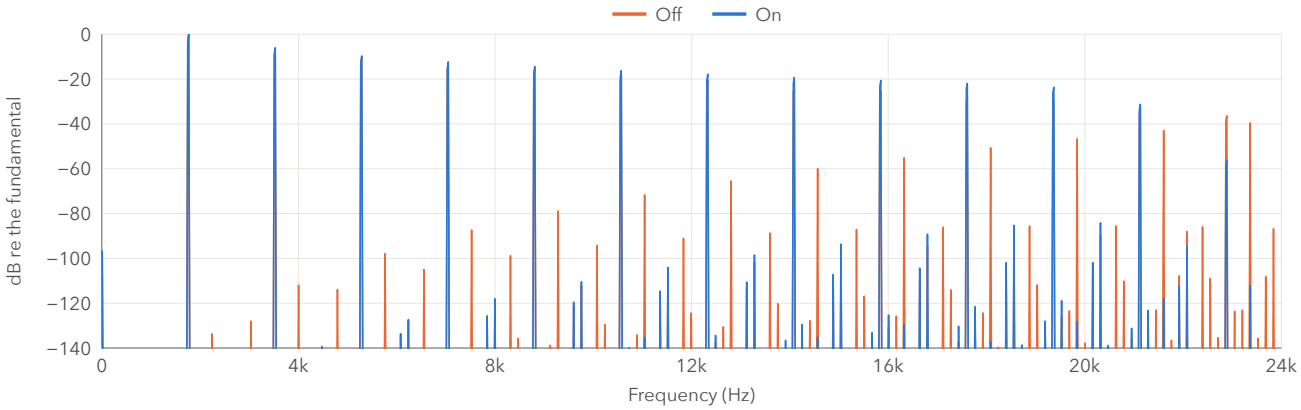


Figure 9. The spectrum of a saw at A6 (1760 Hz) with ANTI-ALIAS off and on. The harmonics are the peaks every 1760 Hz; everything between them is aliasing. The loudest alias below 15 kHz is 60 dB below the fundamental with ANTI-ALIAS off, and 98.5 dB below with it on. The top harmonics are also louder with it on: the harmonic at 17.6 kHz is 22.0 dB down instead of 28.0 (for an ideal saw it is 20.0).

ANTI-ALIAS also keeps more of the highest harmonics. Band-limiting at the host's rate reduces the highest harmonics; at twice the rate, this reduction moves an octave higher, above the audible range. For a 1 kHz saw, the harmonic at 16 kHz is 6.8 dB below an ideal saw's with ANTI-ALIAS off and 1.6 dB below with it on; at 20 kHz, 12.3 dB and 4.1 dB. So the sound is slightly brighter with it on.

ANTI-ALIAS delays the output by 1.5 samples (about 0.03 ms), which only matters if the output is mixed with an undelayed copy of the same pitch. It uses several times as much CPU as the oscillator does at the host's rate. Switching it on or off crossfades between the two over 20 ms, so it does not click.

Notes on the sound

Voices start at different phases. Each voice starts at a different point in its cycle. Two voices playing the same note therefore do not add up to one louder copy of the same wave: their sum has a different, fixed harmonic balance, so a part doubled on two voices sounds like two instruments, even without detuning (Figure 10).

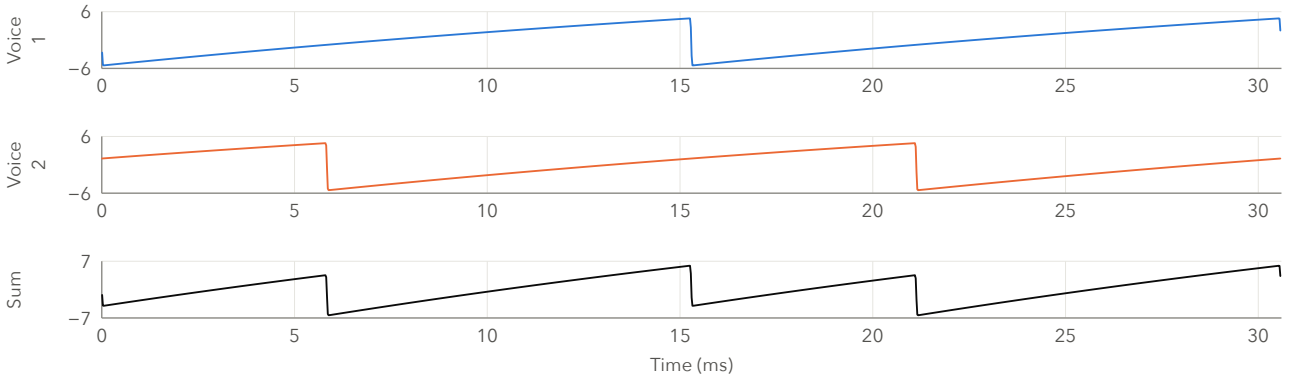


Figure 10. Two voices on the same note, saw only. Voice 2 is 0.618 of a cycle ahead of voice 1, and stays there. Their sum is steady, but its harmonics are changed: compared with one voice, the fundamental is 2.8 dB lower, the second harmonic 3.4 dB higher and the fourth 15.1 dB lower.

A fast pulse-width sweep can cause a slight thump. MIX OUT passes through a 5 Hz high-pass filter, a model of an analogue coupling capacitor, which removes the pulse's DC offset. The DC offset changes with the pulse width, and the filter removes it slowly, so a fast sweep lets some of it through. The same filter makes very low notes slightly quieter and curves their waveforms; this is most audible on the sub below about 40 Hz.

Starting points

Pad. Saw at 10, sub at 4, pulse at 3 with WIDTH just off centre. A slow LFO into the pulse width CV, with AMT about a third of the way up. Play chords through BBD.

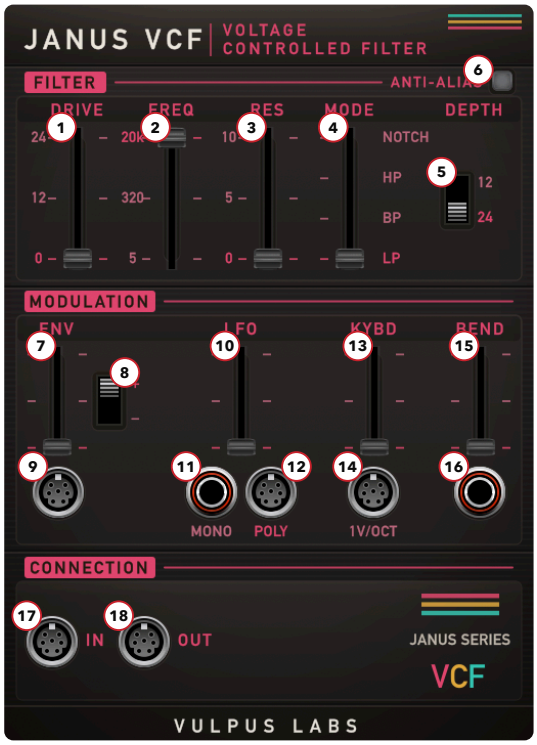
Sync lead. A second oscillator into SYNC; saw fader up, the others down. Modulate this DCO's pitch with an envelope, or turn SEMITONE by hand.

Bell. Saw and pulse down, sub up. A second DCO tuned a fifth higher into PHASE MOD's POLY CV, with a fast-decaying envelope controlling its level. Turn AMT up until the sound becomes metallic.

4 · VCF

VOLTAGE CONTROLLED FILTER

VCF is a four-pole ladder filter for each of up to sixteen voices. By default it is a 24 dB per octave low-pass, set for the level DCO outputs. **DRIVE** adds gain before the filter, **MODE** selects band-pass, high-pass or notch, and **DEPTH** halves the slope. The envelope, LFO, keyboard and pitch bend inputs each move the cutoff, by their own amounts.



1	**DRIVE**	gain into the ladder, 0 to +24 dB
2	**FREQ**	cutoff, 5 Hz to 20 kHz
3	**RES**	resonance, to self-oscillation
4	**MODE**	LP, BP, HP, NOTCH
5	**DEPTH**	24 or 12 dB per octave
6	**ANTI-ALIAS**	runs at four times the sample rate
7	**ENV**	envelope amount
8	**+, −**	envelope polarity
9	**Jack**	envelope, per voice
10	**LFO**	LFO amount
11	**MONO**	LFO, all voices
12	**POLY**	LFO or any CV, per voice
13	**KYBD**	key follow amount
14	**1V/OCT**	pitch, per voice
15	**BEND**	bend amount
16	**Jack**	bend, all voices
17	**IN**	audio, per voice
18	**OUT**	filtered, per voice

How it works

A ladder filter is four one-pole low-pass stages in series, with feedback from the last stage to the input, and a saturating stage inside each pole. This structure gives it three characteristics.

- **Resonance reduces the level.** The feedback is subtracted from the input, so raising RES lowers the level below the cutoff: by 6 dB at a quarter of the way up, and 12 dB at three quarters. VCF does not compensate for this, because compensating would also change the sound.
- **It distorts.** Each stage saturates, so the distortion depends on the input level, the resonance and how far each harmonic is above the cutoff.
- **It compresses loud signals.** With the filter fully open and no resonance, a 0.1 V sine passes at full level (0 dB), but the fundamental of a 4 V sine is 3.5 dB down and of a 5 V sine 4.7 dB down (Figure 11). DRIVE sets how far into this compression a signal goes.

Each voice has its own filter and its own modulation, and the output is still sixteen separate voices. Patch SUM after VCF, and HPF after SUM.

VCF is set up for inputs of ±5 V, which is DCO's output level. A louder input is driven harder even with DRIVE at 0 dB, and a quieter one distorts less. If VCF does not sound as expected, check the input level first.

FILTER

DRIVE

DRIVE adds 0 to +24 dB of gain before the filter. It loads at the bottom, 0 dB, which is the level DCO provides, not an undriven filter. DRIVE affects the input only, not the feedback, so it does not change the resonance. VCF does not reduce its output to compensate, so raising DRIVE makes the output both louder and more distorted. Reduce the level after VCF if needed.

DRIVE	Gain	What it does
bottom	0 dB	clean at normal levels; compresses at full scale
quarter	+6 dB	compresses throughout at DCO's full level
half	+12 dB	audible distortion on every loud note
top	+24 dB	heavy distortion

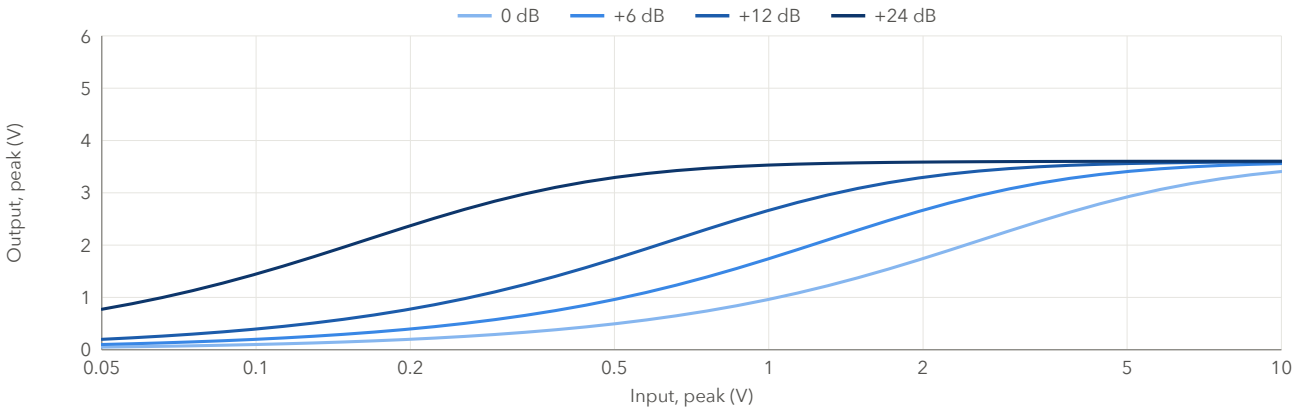


Figure 11. Output level of the fundamental against input level, for a 200 Hz sine with the filter fully open, at four DRIVE settings, measured. All settings approach the same maximum, about 3.6 V. At +24 dB, a 0.1 V input already gives 1.45 V out.

FREQ

FREQ sets the cutoff, over twelve octaves from 5 Hz. **320** on the scale is the geometric midpoint. It loads at the top, where the filter is fully open. The top of the range is 20.5 kHz. At the bottom, 5 Hz, a 1 kHz tone is removed completely, but a 1 Hz control signal still passes.

FREQ	quarter	half	three quarters	top
Cutoff	40 Hz	320 Hz	2.56 kHz	20.5 kHz

RES

RES sets the resonance, from none to self-oscillation. It loads at the bottom.

RES	Peak at the cutoff	With ANTI-ALIAS	Passband
0	−10.9 dB	−11.8 dB	0 dB
0.25	−8.0 dB	−9.2 dB	−6.0 dB
0.5	−3.7 dB	−5.5 dB	−9.5 dB
0.75	+5.5 dB	+1.1 dB	−12.0 dB

Near the top of RES's range the filter oscillates by itself, so the level at the cutoff no longer measures a gain. At a 1 kHz cutoff this starts at about RES 0.88 (0.96 with ANTI-ALIAS on). The level below the cutoff falls to −14.0 dB at the top of RES's range.

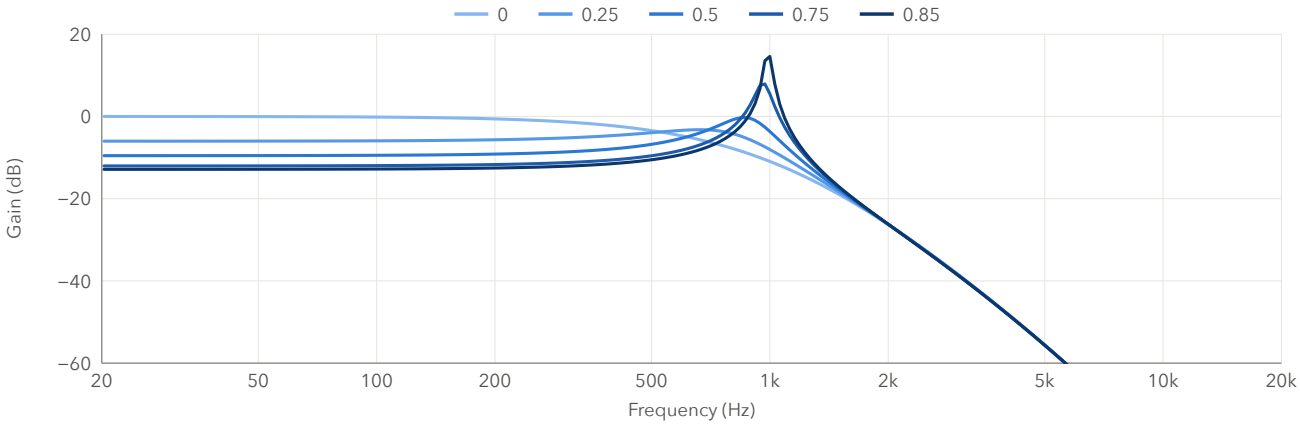


Figure 12. The low-pass at a 1 kHz cutoff, at five RES settings, measured with a small sine. As the peak rises, the level below the cutoff falls. At 0.75 the peak is 7.9 dB up, at 972 Hz, and the level below the cutoff is 12 dB down.

The peak grows slowly at first: at half way it is still below 0 dB, and most of the increase happens in the last quarter of the range. Because the level below the cutoff falls as resonance rises, a resonant bass sound needs its level raised after the filter.

The RES setting at which the filter starts to oscillate depends on the cutoff: higher cutoffs oscillate at lower settings. At 4 kHz the filter oscillates from RES 0.75; at 440 Hz only at the top. The oscillation is a sine with some odd harmonics, and its level rises with frequency: about 0.44 V at 440 Hz, 0.67 V at 1 kHz and 1.43 V at 4 kHz. The oscillation is at the frequency FREQ is set to, so with KYBD at the top, an oscillating VCF can be played from the keyboard.

MODE and DEPTH

MODE has four positions, from the bottom: **LP**, **BP**, **HP** and **NOTCH**. It loads at LP. All four are mixed from the outputs of the same four stages, so they have the same resonance, saturation and loss of level.

DEPTH has two positions: **24** (lower, the default) and **12**. In both, the feedback comes from the fourth stage, so resonance and self-oscillation are the same; only the output is taken from a different stage, which halves the slope. So 12 dB is not a two-pole filter: it keeps the four-pole filter's resonance with a gentler slope.

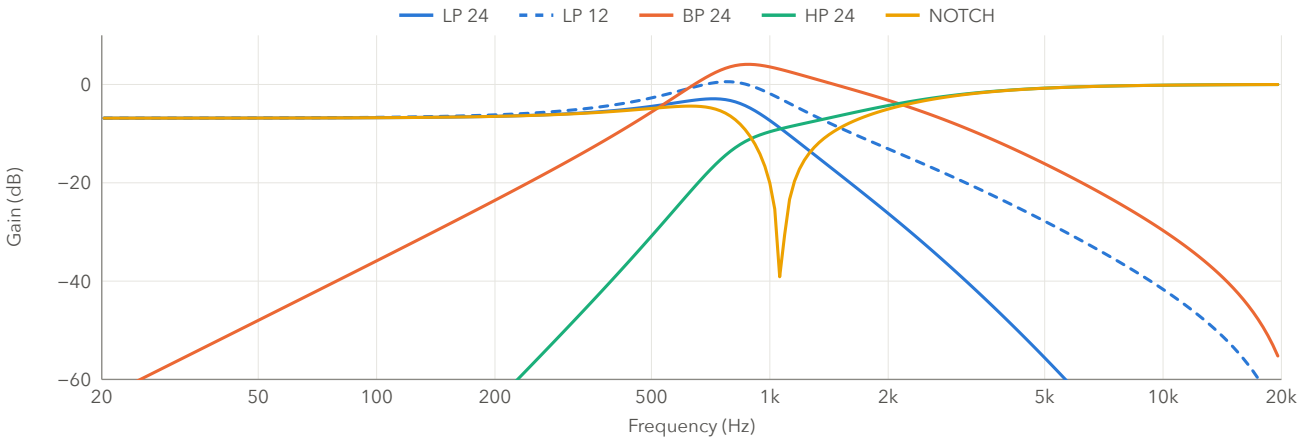


Figure 13. The four modes at a 1 kHz cutoff with RES at 0.3, and the low-pass at both depths, measured. From 2 to 4 kHz, LP 24 falls 21.6 dB per octave and LP 12 half that. The notch's deepest point, −73 dB, is at 1068 Hz, 6.75% above the cutoff, between the plotted points.

The notch's frequency is above the cutoff shown on the scale: 6.7% above at 1 kHz, or 1.7% with ANTI-ALIAS on, so tune it by ear. Keep RES below about 0.5 in NOTCH mode: above that, the resonant peak just below the notch dominates the sound, and the notch itself gets shallower. DEPTH has no effect in NOTCH mode.

Changes of MODE or DEPTH crossfade, settling within about 50 milliseconds, so they do not click, even during a note.

ANTI-ALIAS

ANTI-ALIAS runs every voice's filter at four times the sample rate. It is off by default, and when off, VCF is unchanged.

The saturating stages produce harmonics. At the host's sample rate, harmonics above 24 kHz alias: they reappear at lower frequencies as inharmonic tones, which move in the wrong direction when the pitch changes. At four times the sample rate, these harmonics are removed by a filter before the signal is converted back to the host's rate. ANTI-ALIAS makes the most difference with the filter open and DRIVE up; on a dark, quiet sound there is little aliasing to remove. Turn on DCO's ANTI-ALIAS as well.

Inharmonic energy below 15 kHz	Note	Off	On
Saw, filter open	A4	−39.5 dB	**−74.4 dB**
	C6	−35.8 dB	**−70.4 dB**
Saw, open, DRIVE +12 dB	A4	−29.9 dB	**−72.5 dB**
	C6	−26.1 dB	**−68.5 dB**

At four times the sample rate the filter is closer to an analogue ladder, and sounds slightly different:

- the resonance is lower: at RES 0.75 the level at the cutoff is +1.1 dB instead of +5.5 dB (Figure 14);
- self-oscillation is within 1.5 cents of FREQ up to 4 kHz (with ANTI-ALIAS off it is 26 cents sharp at 4 kHz), and about half as loud;
- fully open, the highest frequencies are slightly reduced: −1.8 dB at 10 kHz and −4 dB at 15 kHz;
- the notch is closer to the cutoff shown on the scale.

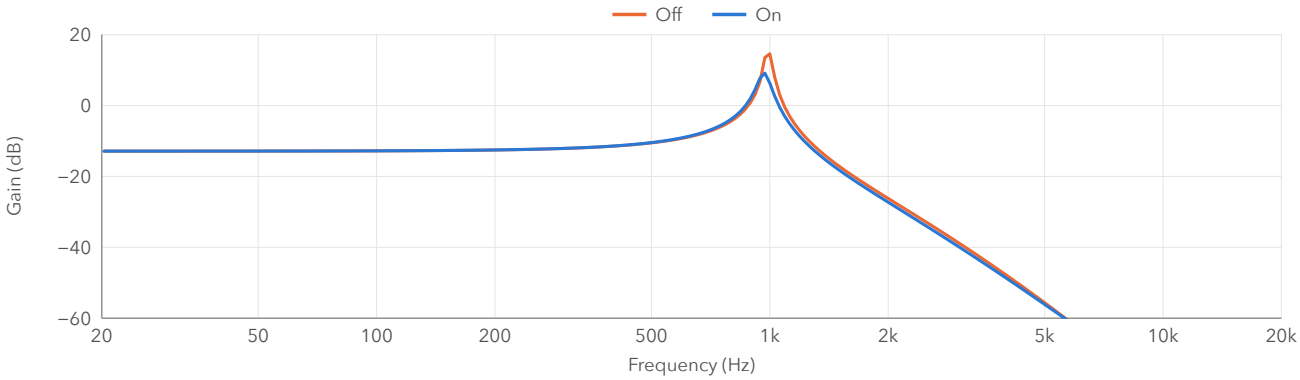


Figure 14. The low-pass at 1 kHz with RES at 0.85, ANTI-ALIAS off and on, measured. The peak is 14.6 dB up with it off, and 9.1 dB up with it on, slightly below the cutoff. The level below the cutoff is the same.

ANTI-ALIAS delays the signal by 0.25 ms, which only matters if the output is mixed with a parallel path that does not have ANTI-ALIAS on. It uses about ten times as much CPU as the filter at the host's rate. Switching it on or off crossfades over 20 ms; during the crossfade the two signals are 0.25 ms apart, which can be heard as a brief whooshing sound.

MODULATION

There are four amount sliders, each above the input it scales. Each runs from zero to full, and none inverts. **All the modulation inputs are added together in octaves**, so two inputs that each raise the cutoff by an octave raise it by two octaves, wherever FREQ is set.

ENV has the widest range: at full, a 5 V envelope moves the cutoff by 14 octaves, enough to reach either end of FREQ's range from any setting. Most patches need less than half. The **+** and **−** switch inverts the envelope at the same depth. It loads at +. The input is polyphonic, so each voice's filter follows its own envelope.

LFO moves the cutoff up to 3.5 octaves either way at full. The **MONO** input takes one LFO for all voices; the **POLY** input takes a separate value for each voice, from LFO's per-voice outputs or any other poly CV. Use POLY as VCF's per-voice cutoff CV input; for example, patch a fan through CVC into it (chapter 11).

KYBD sets keyboard tracking. At the top, the cutoff follows the 1 V/oct pitch at the **1V/OCT** input exactly. Lower settings track partially, which stops high notes becoming too bright without making low notes dull.

BEND moves the cutoff up to 3.5 octaves either way at full, from a mono input.

There is no separate cutoff CV input: use LFO POLY. Modulation is read at 1.5 kHz and ramped between readings, which is fast enough for envelopes and LFOs, but not for audio-rate filter FM.

Starting points

Classic filter. DRIVE at the bottom, LP, 24. FREQ near the middle, RES at two thirds, ENV at a third on +, KYBD at half.

Resonant bass. FREQ low, RES at 0.85, ENV low with a fast-decaying envelope. Raise the level after the filter.

Sine oscillator. RES and KYBD at the top, and the pitch CV into 1V/OCT. The filter does not start oscillating from silence: patch a short pulse, such as KEY's TRIG, into IN to start it, and it keeps going. FREQ sets the transposition.

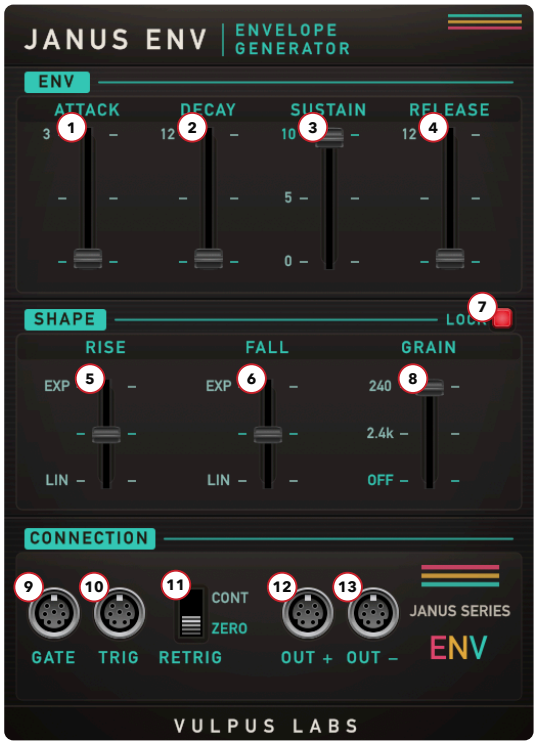
Band-pass sweep. BP, 12, RES about 0.6, an LFO at moderate depth: a vocal, wah-like sound.

Moving notch. NOTCH, RES at 0.3 or below, a slow LFO at a third: a phaser-like sweep on a bright sound.

5 · ENV

ENVELOPE GENERATOR

ENV is an ADSR envelope for each voice. As well as a gate input, it has a **TRIG** input, which restarts the envelope while the gate is already high. **RISE** and **FALL** set the curves of the stages, and **GRAIN** updates the output in steps, at a rate you set.



1	**ATTACK**	1.5 ms to 3 s
2	**DECAY**	1.5 ms to 12 s
3	**SUSTAIN**	0 to 10
4	**RELEASE**	1.5 ms to 12 s
5	**RISE**	curve of the attack
6	**FALL**	curve of the decay and release
7	**LOCK**	links FALL to RISE
8	**GRAIN**	output update rate
9	**GATE**	high: attack, decay, sustain; low: release
10	**TRIG**	restarts the envelope
11	**RETRIG**	ZERO or CONT
12	**OUT +**	0 to +5 V
13	**OUT –**	0 to −5 V

Stage times

Each stage takes its set time, wherever it starts and ends. Many envelopes do not work this way. In ENV:

- **DECAY does not depend on SUSTAIN.** With DECAY at 2 s, the decay takes 2 seconds whether it falls to 0 or to 9.5.
- **A release from a low level takes the full release time.** A note released from a sustain of 1 takes as long to fade as one released from 10.
- **A stage with no distance to cover is skipped.** With SUSTAIN at the top, there is no decay.
- **A retriggered attack takes the full attack time**, from wherever RETRIG starts it.

Every stage reaches its target level exactly at the end of its time.

GATE, TRIG and the outputs

GATE is high when above 0 V. While it is high, the envelope runs attack, decay and sustain; when it goes low, the release starts. If the gate goes low during the attack, the release starts from the level reached.

TRIG restarts the envelope on a rising edge while the gate is high; during the release, a trigger restarts the release from the current level. A one-sample pulse is enough. Patch KEY's TRIG output here: when KEY gives a voice a new note while its gate is still high, there is no new gate edge, and without TRIG the envelope would not restart. A gate rising and a trigger in the same sample start one attack, not two. If TRIG is not patched, it has no effect.

OUT + outputs 0 to +5 V, the range VCF's ENV input and SUM's ENV input expect. **OUT –** outputs the same envelope inverted, 0 to −5 V, so it falls when a note starts. Both outputs are always active, so one envelope can open one thing and close another.

ENV

Slider	Range	Default
ATTACK	1.5 ms to 3 s	bottom
DECAY	1.5 ms to 12 s	bottom
SUSTAIN	0 to 10, in tenths of full scale	top
RELEASE	1.5 ms to 12 s	bottom

The panel shows the top of each slider's range; the tooltip shows the time, in milliseconds below one second. The time sliders are exponential, so the same slider movement gives the same ratio anywhere on the slider. The middle of the slider is 67 ms for ATTACK and 134 ms for DECAY and RELEASE: the bottom half of each slider covers the short times, and the top half the long ones. The shortest time, 1.5 ms, is fast but does not click.

SUSTAIN takes effect immediately. Moving it while notes are held changes their level. Moving it during a decay changes the decay's target without changing when the decay ends. The tooltip shows the level both ways, for example `7.0 of 10 (3.50 V)`. **The times are fixed when each stage starts**: moving ATTACK during an attack changes the next attack, not the current one.

SHAPE

RISE and FALL

RISE sets the curve of the attack. **FALL** sets the curve of the decay and the release. Each runs from **LIN** (a straight line) at the bottom to **EXP** (the most curved) at the top, and loads at its centre (Figure 15). Changing a curve does not change any stage's time. You can move the curves while a note is held; the level changes, but the stage timing does not.

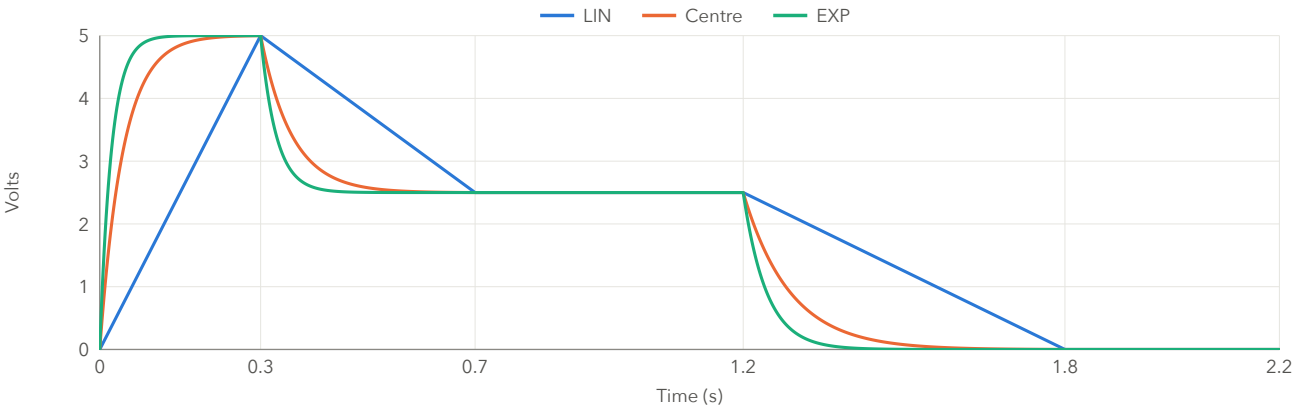


Figure 15. One envelope, ATTACK 0.3 s, DECAY 0.4 s, SUSTAIN 5, RELEASE 0.6 s, with the gate going low at 1.2 s, and RISE and FALL locked together at three settings, measured. Each stage ends at the same time for all three. Halfway through the attack, LIN is at 2.5 V, the centre setting at 4.85 V and EXP at 5.0 V.

LOCK, the small button at the right end of the SHAPE section, links the two curves: when you turn it on, FALL is set to RISE's value, and moving either one moves both. It is on by default. Turn it off to set different curves, for example a linear attack with a curved release.

GRAIN

GRAIN sets how often the envelope's output is updated. Below every-sample updating, the output holds its value between updates, so a fast envelope rises and falls in steps, like an envelope computed by a microprocessor and written to each voice periodically.

GRAIN	Update rate	Updated every
bottom, **OFF**	every sample (smooth)	sample
middle, **2.4k**	2.39 kHz	20 samples at 48 kHz

top, **240**	238 Hz	202 samples (4.2 ms)

The numbers on the panel decrease as the slider goes up, because a higher setting means a lower update rate. The tooltip shows both the rate and the interval, for example `2.39 kHz (every 20 samples)`.

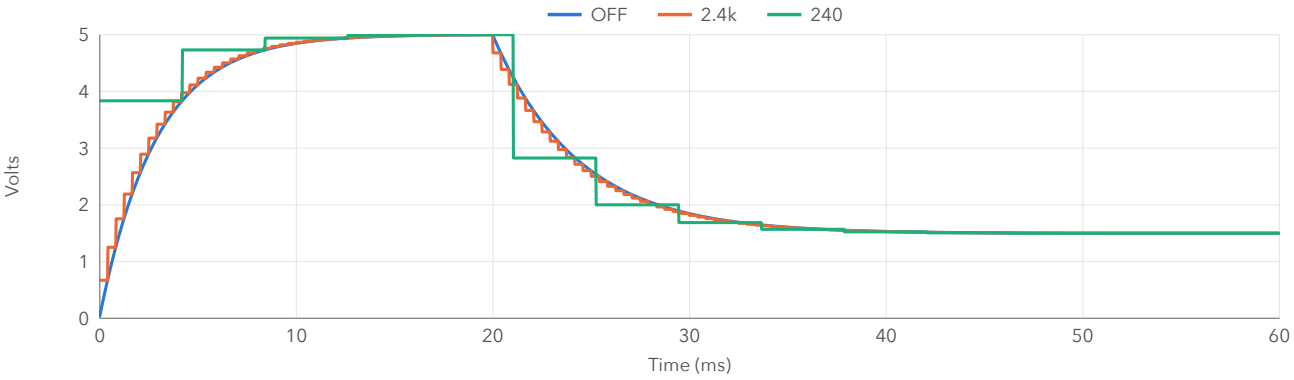


Figure 16. The first 60 ms of a fast envelope, ATTACK 20 ms and DECAY 30 ms to a SUSTAIN of 3, at three GRAIN settings, measured. At 240 the attack is five steps, and the first step is 3.8 V. Here the first step happens on the same sample as the gate; in general it can come up to one update interval later.

GRAIN has no audible effect on a slow envelope. The step size is how far the level moves between updates: at 240, a 1.5 ms attack is a single step, but a one-second attack moves at most about 3% of full scale per step (about 6% with RISE at EXP). The steps are most audible on fast envelopes controlling something sensitive, such as a filter cutoff, rather than a VCA.

Each stage starts on the sample the gate or trigger edge arrives, but the output changes only at the voice's next update, so with GRAIN up the start can be heard up to one update interval late. Stage times are not lengthened either, but a stage can end up to one update interval early. The voices' updates are spread across the interval rather than all happening at once. Moving GRAIN re-spreads them, which causes a small step on every voice, so set GRAIN and leave it rather than moving it while notes are held.

RETRIG

RETRIG sets what a trigger does to a voice whose envelope is not at zero (Figure 17).

- **ZERO** (down, the default): the envelope falls to zero, then starts the attack. The fall is a straight line, taking 2 ms from full scale and less from lower levels, which is long enough not to click. The fall is not counted in the attack time.
- **CONT**: the attack starts from the current level, with no fall.

ZERO is the default because with CONT a retrigger is often inaudible: at a sustain of 10 the attack has nowhere to rise to. Use CONT when the envelope is retriggered from a low level, or when ZERO's dip is unwanted. A change of RETRIG applies to the next trigger.

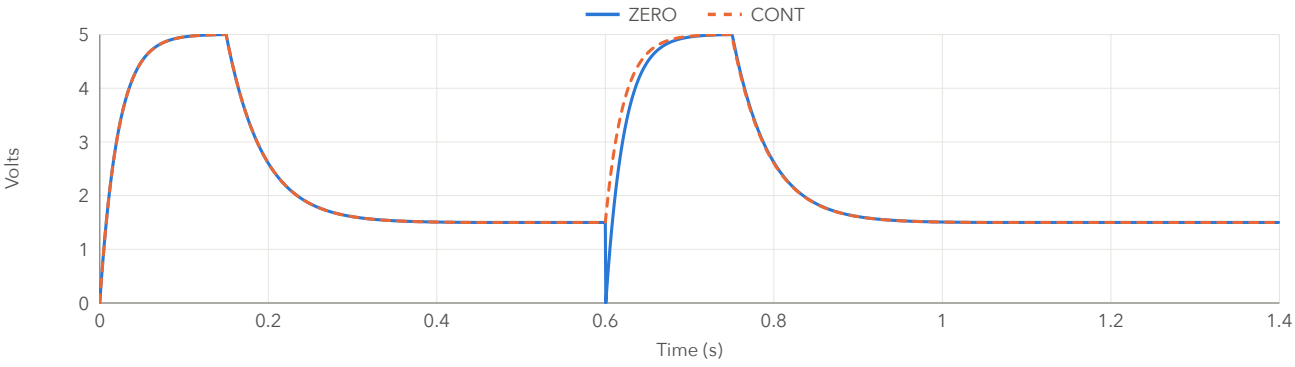


Figure 17. A held note, ATTACK 0.15 s, DECAY 0.3 s, SUSTAIN 3, and a trigger at 0.6 s, measured. ZERO falls from 1.5 V to 0 V in 0.6 ms, then runs the full attack; CONT starts the attack from 1.5 V. Both reach the top at about 0.75 s.

Starting points

Classic envelope. LOCK on, both curves at their centre, GRAIN at 240, RETRIG at ZERO.

Pad. ATTACK about a second, DECAY long, SUSTAIN about 7, RELEASE one to two seconds. OUT + to VCF's ENV input with the ENV slider at a third, and to SUM's ENV input.

Plucked bass. ATTACK at the bottom, DECAY about 200 ms, SUSTAIN 0, RELEASE short. FALL well towards EXP. GRAIN at 240 adds steps to the start of every note.

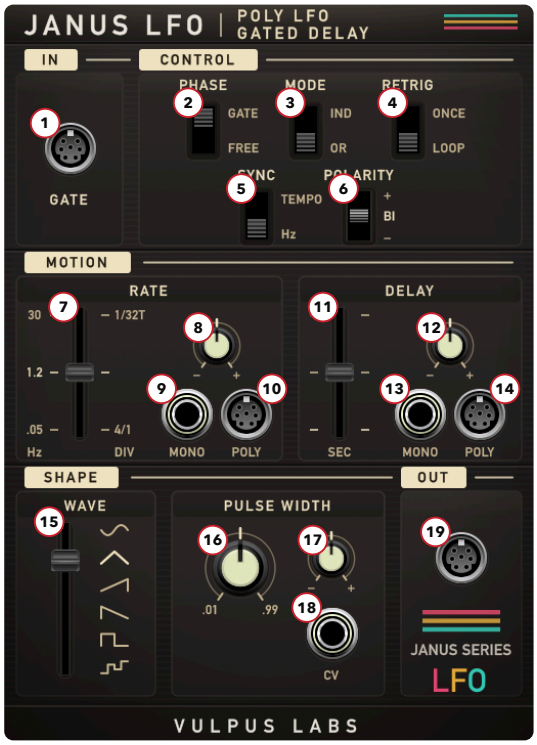
Retriggered voices. KEY's ORDER at ROTARY, KEY's GATE to GATE and KEY's TRIG to TRIG. Hold a chord and play other notes: every voice given a new note starts a new envelope.

An envelope and its inverse. OUT + to VCF's ENV input, and OUT − to a second filter's cutoff. (BBD's MIX CV is a mono input, so it would read only the first voice's envelope.)

6 · LFO

POLY LFO · GATED DELAY

LFO is an LFO with up to sixteen lanes. Every lane has the same waveform, but a poly CV can set a different rate and fade-in time for each lane. A gate can restart each lane's fade-in and its waveform, either for each voice separately or once per phrase. With only **OUT** patched, it is an ordinary LFO, with every lane identical.



1	**GATE**	poly
2	**PHASE**	FREE, or restarted by the gate
3	**MODE**	OR: per phrase; IND: per voice
4	**RETRIG**	LOOP, or ONCE per trigger
5	**SYNC**	RATE in Hz, or in tempo divisions
6	**POLARITY**	+, BI or −
7	**RATE**	0.05 to 30 Hz
8	**AMT**	rate CV amount
9	**MONO**	rate CV, all lanes, ±1 octave
10	**POLY**	rate CV, per lane
11	**DELAY**	fade-in time, 0 to 5 s
12	**AMT**	delay CV amount
13	**MONO**	delay CV, all lanes
14	**POLY**	delay CV, per lane
15	**WAVE**	six waveforms
16	**PULSE WIDTH**	pulse duty cycle
17	**AMT**	pulse width CV amount
18	**CV**	pulse width CV
19	**OUT**	poly, ±5 V

GATE

GATE takes a gate for each voice, high when above 0 V. With nothing patched, the LFO runs continuously at full depth, with every lane identical, and DELAY, PHASE and MODE have no effect. Patch KEY's GATE output here to use them.

The five switches

Every switch loads in the down position, except PHASE, which loads at GATE (up), and POLARITY, which loads in the middle. With nothing patched into GATE, the defaults give a classic LFO: one continuous waveform, shared by every note.

PHASE sets whether a gate restarts the waveform.

- **FREE** (down): gates do not affect the waveform; each note starts wherever the waveform is.
- **GATE** (up, the default): the waveform restarts when a note starts. Sine and triangle restart at 0 V, rising. The saws, the pulse and random restart at the start of their cycle: saw up at −5 V, saw down at +5 V, the pulse on its high half, and random with a new value that lasts a whole cycle. Under ONCE every waveform restarts at the start of its cycle. MODE sets what counts as a note starting.

MODE sets whether the gate acts per phrase or per voice.

- **OR** (down): all lanes share one fade-in and one waveform. They start when the first gate rises after all gates were low. Adding a note to held notes does not restart anything, so a note played later in a phrase starts at whatever depth the fade-in has reached.
- **IND**: each voice's gate starts its own fade-in and, with PHASE at GATE, restarts its own waveform. A note played later in a phrase fades in from its own start. With PHASE at GATE, the lanes restart at different times, so they no longer move together; with PHASE at FREE they stay in step and only their depths differ.

Switching from IND to OR during a fade-in does not cause a jump: the lanes' depths converge over the delay time, and their phases come together at the start of the next phrase (if PHASE is at GATE).

RETRIG sets **LOOP** (down), where the waveform repeats, or **ONCE**, where each trigger plays one cycle and then holds. With PHASE at GATE, a ONCE cycle runs from the beginning of the waveform to its end, and holds the last value. On BI, saw down falls from +5 V to −5 V and stays there; saw up rises from −5 V to +5 V and stays there; the pulse starts high, goes low at the point PULSE WIDTH sets, and stays low. With POLARITY at +, saw down is a single fall from full to zero, a simple envelope (Figure 20). With PHASE at FREE, a ONCE cycle starts and stops wherever the waveform was, and holds that value until the next trigger. With nothing patched into GATE, switching RETRIG to ONCE triggers one cycle.

SYNC sets whether the RATE slider is in hertz, **Hz** (down), or in note divisions of the tempo, **TEMPO**.

POLARITY sets the output range.

POLARITY	Output	Range
+	(x + 1) / 2	0 to +5 V
BI	x	−5 to +5 V
−	−(x + 1) / 2	−5 to 0 V

The + and − settings give the same waveform at half the peak-to-peak level, so an LFO on + into a filter cutoff only raises the cutoff above the slider's setting. With a fade-in, the output on + or − fades in from 0 V.

MOTION

RATE

The **RATE** slider sets the rate from 0.05 to 30 Hz, exponentially. It loads at the middle, about 1.2 Hz. With SYNC at TEMPO, the slider selects one of twenty note divisions, from 4/1 at the bottom to 1/32T at the top, which is 0.125 to 24 Hz at 120 BPM. The tooltip shows the rate in hertz, and the division when SYNC is at TEMPO.

MONO and **POLY** are added together and scaled by **AMT**. At full AMT, ±5 V changes the rate by an octave either way: up to double or down to half. The CV multiplies the rate, so the same CV has the same relative effect wherever the slider is, including in TEMPO mode. A CV into MONO changes the rate of every lane; a fan from KEY into POLY gives each lane of a stack a different rate (Figure 3).

DELAY

The **DELAY** slider sets a fade-in time, from 0 to 5 seconds (it loads at 2.5 s), during which the depth rises from zero to full. MODE sets what starts the fade-in. At 0 there is no fade-in. **MONO**, **POLY** and **AMT** modulate it as they do RATE; at full AMT, ±5 V covers the slider's whole range. A fan into POLY gives each voice a different fade-in time.

The depth never decreases. It rises while the gate that started the fade-in is high, stays at full depth once it gets there, and stops rising if the gate goes low. It is not an envelope: notes released during the fade-in keep their depth through their release, and the next phrase starts the fade-in again from zero (Figure 18).

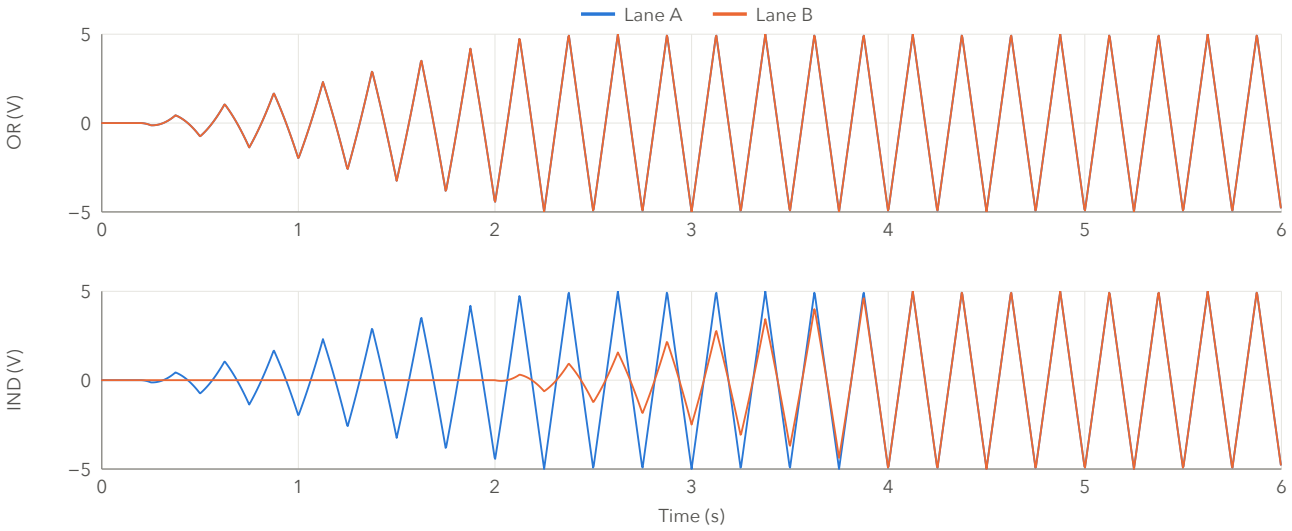


Figure 18. Two lanes, a 4 Hz triangle, DELAY 2 s, PHASE at FREE, measured. Lane A's gate rises at 0.2 s and lane B's at 2 s, and both fall at 5 s. With MODE at OR, lane B starts at the shared fade-in's current depth, 90%. With MODE at IND, lane B fades in over 2 s from its own gate. After the gates fall, both lanes keep their depth.

The poly CVs for RATE and DELAY are read at 1.5 kHz, which is fast enough for envelopes, LFOs and fans. A mono input is read every sample while its POLY input is empty, and at 1.5 kHz when both are patched.

SHAPE

WAVE selects one of six waveforms, from the top: **sine**, **triangle**, **saw up**, **saw down**, **pulse** and **random**. It loads at triangle. Random is a sample-and-hold: it takes a new random value each time the waveform restarts, at the end of each cycle and when a gate restarts it. Lanes that restart on the same sample get the same value. Under OR all lanes have the same value; under IND with PHASE at GATE, notes started at different times have different values.

The waveforms are not smoothed: the saws, pulse and random have instant steps. To smooth them, patch SLU after LFO.

PULSE WIDTH sets the pulse's duty cycle, with a square at its centre, and has its own **AMT** and **CV**. A pulse restarted by a gate is high at the start of the note, at any width.

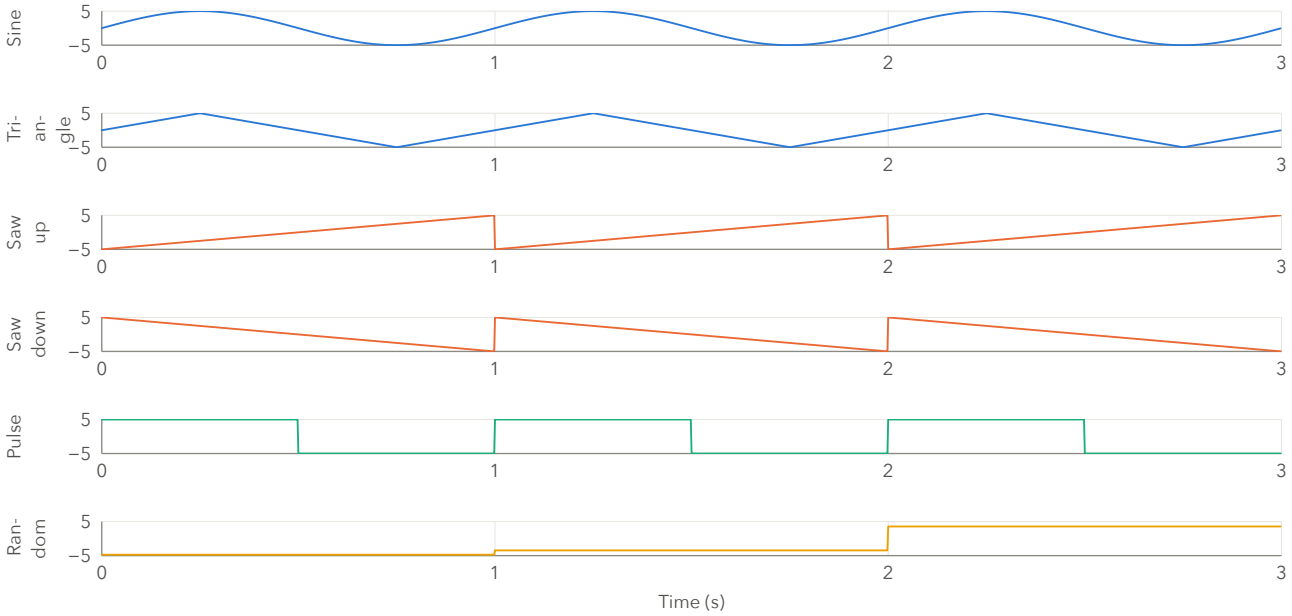


Figure 19. The six waveforms at 1 Hz, on BI, DELAY 0, measured, with a gate at 0 s restarting each waveform: sine and triangle at 0 V rising, the others at the start of their cycle. Random takes a new value each time the waveform restarts.

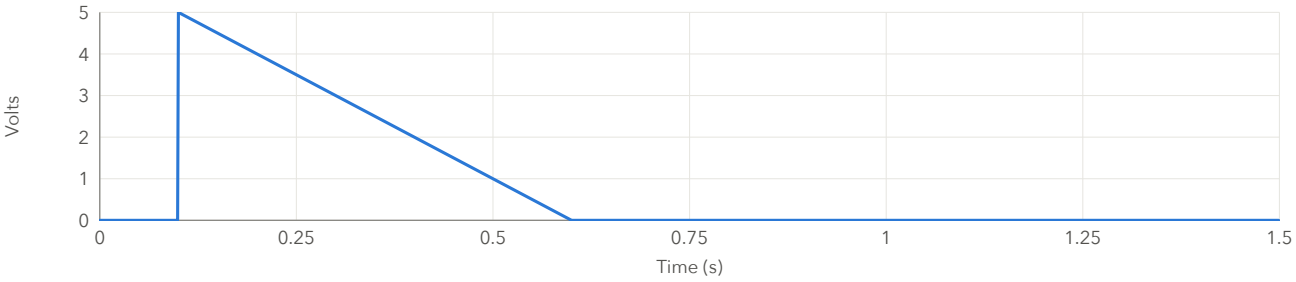


Figure 20. A one-shot, measured: saw down at 2 Hz, PHASE at GATE, RETRIG at ONCE, POLARITY at +, DELAY 0, with the gate rising at 0.1 s. The output falls from 5 V to 0 V over half a second, then holds.

OUT

OUT outputs one value per lane, ±5 V at full depth on BI. Under OR every lane has the same value. The tooltips show the slider and mono CV values only, not the per-lane values.

Starting points

Vibrato. Only OUT patched, to DCO's FM input at a small amount. Triangle, RATE at its centre, BI.

Delayed vibrato. KEY's GATE into GATE, MODE at OR, DELAY about a second. The vibrato fades in once per phrase, and stays at full depth while any note is held.

Fade-in per note. As above, with MODE at IND and PHASE at FREE: each note's vibrato fades in from the start of that note.

One-shot envelope. PHASE at GATE, RETRIG at ONCE, POLARITY at +, saw down, RATE slow, MODE at IND, DELAY at 0: each note gets one fall from full to zero.

Stack with drifting vibrato. Any of the above, with KEY's LIN fan output into RATE's POLY input and AMT a little right. The lanes of each stack run at slightly different rates, so they drift apart and back together.

7 · SLU

SLEW LIMITER

SLU limits how fast each lane of a signal can change, with separate limits for rising and falling. On a pitch CV it gives portamento; on a gate, an attack and release; on a stepped LFO, ramps between the steps. With both faders at the bottom, SLU is completely transparent and has no effect.



1. **RISE** time to rise 10 V, 0 to 30 s
2. **LIN, EXP** shape of the rise
3. **AMT** rise CV amount
4. **MONO** rise CV, all lanes
5. **POLY** rise CV, per lane
6. **LOCK** links the two faders
7. **FALL** the same, for falling
8. **IN** poly
9. **OUT** poly

What it does

The output follows the input, but cannot change faster than the set rate. If the input changes more slowly than that, the output equals the input. If the input changes faster, or jumps, the output moves towards it at the maximum rate and arrives later. The effect depends on what you patch in:

- a **pitch CV**: jumps between notes become glides (portamento);
- a **gate**: the on and off edges become a rising and a falling ramp, an attack and a release;
- a **stepped signal**, such as a random or square LFO: the steps become ramps;
- a **coarse control**, such as a MIDI controller or automation: its steps become smooth;

RISE sets the limit for a rising signal and **FALL** for a falling one, so, for example, a signal can rise quickly and fall slowly. Each lane is processed separately. When the module loads, or the patch's number of voices goes up, each new lane's output starts at its input value, rather than gliding up from 0 V. A cable patched into IN on a running SLU glides from 0 V, because the empty input was read as 0 V.

TIME

Each fader sets a time from 0 to 30 seconds, and loads at 1 second. **The time is how long the output takes to change by 10 V**, so it sets a rate, not a fixed duration: a change of 5 V takes half as long. Most changes in a patch are small. At 1 V per octave, a semitone is 1/12 V.

Fader	10 V	One octave	One semitone
30 s, the top	30 s	3 s	250 ms
1 s, the default	1 s	100 ms	8.3 ms
100 ms	100 ms	10 ms	0.83 ms

10 ms	10 ms	1 ms	0.083 ms

So settings for glides between notes are near the top of the fader, and settings for smoothing are near the bottom. The fader is not linear: the bottom half covers 0 to 1 second and the top half 1 to 30 seconds. Use the tooltip to read the exact time. **At the bottom the time is exactly zero, and SLU is completely transparent.**

SHAPE

LIN (down, the default) moves at a constant rate and stops when it reaches the input, so the move ends with a corner. **EXP** moves quickly at first and slows down as it approaches the input, so the move ends smoothly. Both take the same time to reach the input (Figure 21), and you can switch between them during a move without a jump. On a pitch CV, EXP gives the glide most players expect; on a gate, EXP gives the curved attack and release of an analogue envelope. There is one switch for each direction.

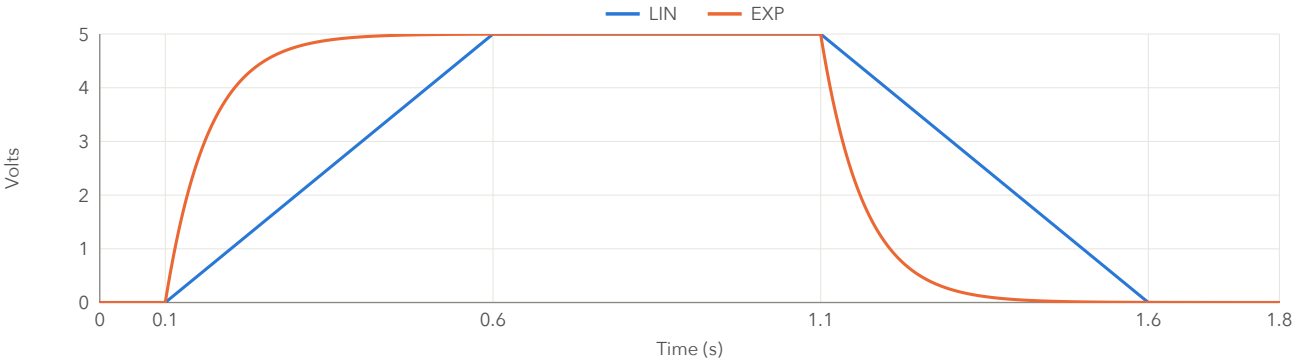


Figure 21. A step from 0 to 5 V at 0.1 s and back at 1.1 s, with both faders at 1 s, measured. A 5 V change takes 0.5 s in either shape: LIN arrives exactly at 0.5 s, and EXP is within 2.5 mV by then and settles the remainder after.

CV

Each direction has **AMT**, **MONO** and **POLY**. The two inputs are added together and scaled by AMT, which has no effect at its centre. **Each volt doubles the time, and each negative volt halves it:** at full AMT, ±5 V multiplies or divides the time by 32, within the 0 to 30 s range. So the same CV has the same relative effect wherever the fader is set, and the CV alone cannot take the time to zero, although a time shorter than one sample acts as zero. With the fader at zero the output always equals the input, whatever the CV.

AMT can be set negative to invert the CV. For example, with one envelope patched into both directions and opposite AMT settings, the rise gets faster while the fall gets slower. MONO affects every lane; POLY takes a separate value for each lane, so a fan from KEY gives each voice of a stack a different glide time (Figure 30).

LOCK, the small button by the FALL fader, links the two faders, so moving one moves the other. Turning LOCK on moves FALL to match RISE. It only links the faders: each direction still has its own CV inputs.

Starting points

Portamento. KEY's V/OCT into IN, and OUT to DCO's V/OCT. LOCK on, both faders well up, both switches at EXP.

Different glide per voice. As above, with a fan from KEY into the RISE and FALL POLY inputs and AMT a little right. The voices of a chord reach the new notes at different times.

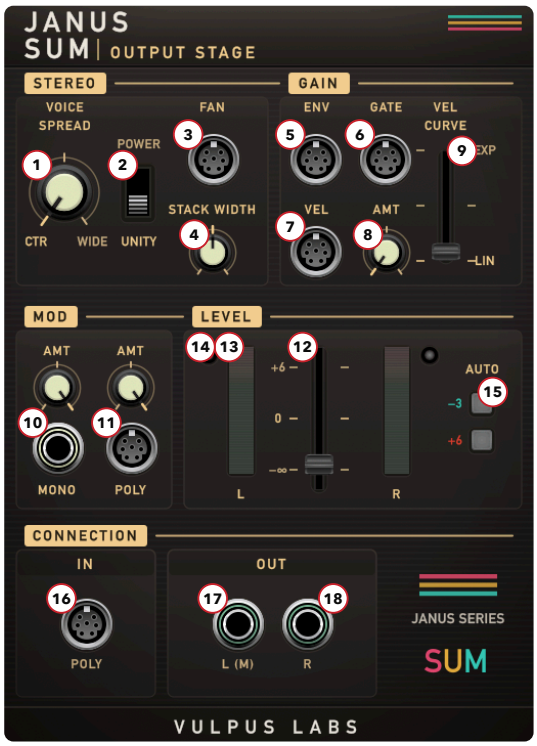
Envelope from a gate. KEY's GATE into IN, RISE a few milliseconds, FALL one second, both at EXP, and OUT to a VCA input such as SUM's ENV input.

Smoothed steps. LFO's OUT, with WAVE at random or pulse, into IN, both faders at 10 to 50 ms, both at EXP.

8 · SUM

OUTPUT STAGE

SUM is the output stage. It takes up to sixteen voices on a poly cable and mixes them to a stereo pair. Each voice passes through a VCA, controlled by an envelope, a gate and velocity, then through a tremolo stage, and is placed in the stereo field, either by lane number or by a fan input. With only the input and outputs patched and the controls where they load, the **L (M)** output is exactly the mono sum of the voices.



1	**VOICE SPREAD**	spreads lanes across the stereo field
2	**UNITY, POWER**	the pan law
3	**FAN**	stereo position per voice
4	**STACK WIDTH**	scales the FAN input
5	**ENV**	envelope, per lane
6	**GATE**	gate, per lane
7	**VEL**	velocity, per lane
8	**AMT**	velocity's amount
9	**VEL CURVE**	LIN to EXP
10	**MONO, AMT**	tremolo, all lanes
11	**POLY, AMT**	tremolo, per lane
12	**Fader**	output level, −∞ to +6 dB
13	**Meters**	L and R, peak
14	**Lamps**	OVER, on each leg
15	**AUTO**	sets the fader for a −3 or +6 dB peak
16	**IN**	poly
17	**L (M)**	left, or the mono sum
18	**R**	right

STEREO

VOICE SPREAD places the voices across the stereo field by lane number. At **CTR**, where it loads, every lane is in the centre, and both outputs carry the same mono sum. Turned towards **WIDE**, the lanes are spread evenly from left to right (Figure 22), across the patch's number of voices, so with sixteen voices and four notes held, the four notes are all to the left. The position depends on the lane, not the note, so a voice's position does not change from one chord to the next.

FAN takes a stereo position for each voice from another module, usually one of KEY's FAN outputs, so that each key's stack is spread from left to right around the centre. **When a cable is patched into FAN, it replaces VOICE SPREAD**, and VOICE SPREAD is greyed out. **STACK WIDTH** scales the FAN input: fully clockwise, a fan from −5 V to +5 V is spread from hard left to hard right; at half, half as wide; left of centre, the positions are reversed. At its centre, where it loads, every voice is in the centre. Unplugging FAN returns control to VOICE SPREAD.

LAW sets how a lane's position is converted to left and right gains.

- **UNITY** (down, the default): a lane in the centre has full gain on both outputs, so the L output alone carries the whole mono sum at full level.
- **POWER**: constant power, so a voice is equally loud anywhere in the field; a lane in the centre has a gain of 0.707 on each output.

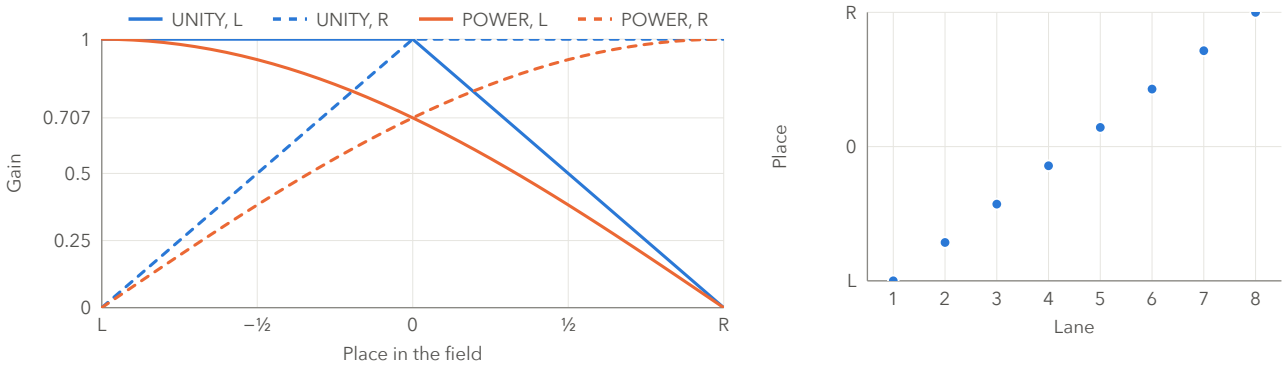


Figure 22. Left: the two laws, measured on one lane moved across the field. Right: the positions of eight lanes with VOICE SPREAD at WIDE, evenly spaced from left to right; with an even number of lanes, none is in the centre.

GAIN

This section is the per-voice VCA. **ENV**, **GATE** and **VEL** can all be used at once, and their effects are multiplied together.

Patched	What a lane gets
nothing	gain 1: SUM only mixes
ENV only	the envelope's level
GATE only	full level while the gate is at 5 V, silence at 0 V
both	the envelope's level while the gate is at 5 V, silence at 0 V

ENV and GATE have no amount knobs, because a VCA's CV attenuated below full scale cannot close the VCA fully, and voices would still be heard after their envelopes closed.

VEL, with **AMT** and **VEL CURVE**, scales each lane by its velocity. AMT loads fully anticlockwise, where velocity has no effect. VEL CURVE runs from LIN to EXP and shapes the velocity response only (Figure 23). The shape of the envelope is set on ENV.

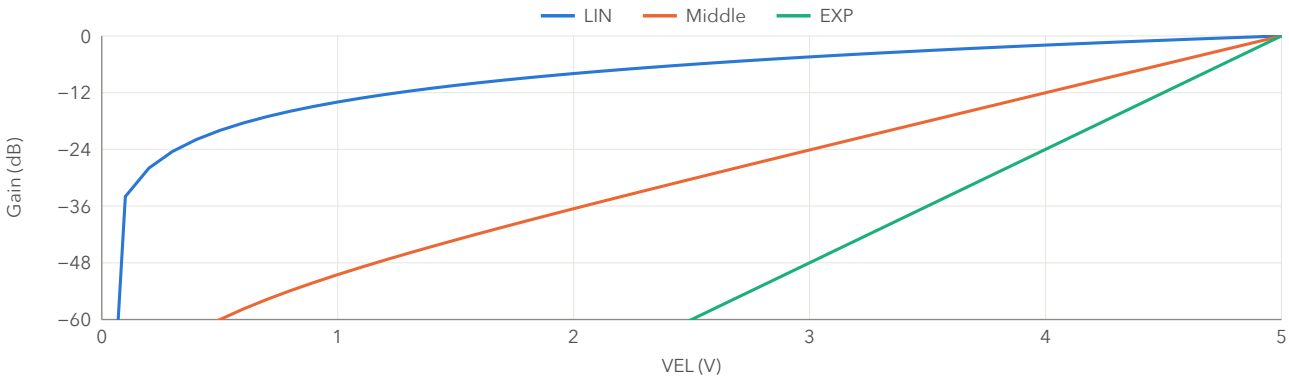


Figure 23. Gain against velocity, with AMT fully clockwise, at three VEL CURVE settings, measured. At 2.5 V (half velocity), the middle setting is 30 dB down and EXP 60 dB down, so the curve increases the difference between soft and hard notes.

Smoothing. Each lane's total gain passes through a one-pole low-pass filter of about one millisecond (near 160 Hz), so a gate opens over about a millisecond rather than instantly, and a stepped envelope is slightly smoothed. There is no control for it.

MOD

The MOD section is a tremolo stage, after the VCA. The **MONO** and **POLY** inputs, each with its own **AMT**, are added together, limited to ±1, and multiplied into every lane. Both AMTs load fully clockwise. A value of −1 gives full level with the phase inverted, so a bipolar LFO gives through-zero tremolo. Beyond full scale in either direction, the gain is limited rather than increased. With neither input patched, the stage is bypassed.

LEVEL

The fader sets the level of both outputs: gain 1 (0 dB) at the middle, and 2 (+6 dB) at the top. It is the last stage before the outputs.

The meters, L and R, show each output's peak level. They rise immediately and fall over about a third of a second. **The OVER lamps** beside them light for one second when an output goes beyond full scale (±5 V). SUM does not clip the output; the lamp shows that the level is higher than the rest of the series expects.

The two AUTO buttons set the fader so that the peak level of the last few seconds would reach a target, and then leave the fader there for you to adjust. **−3** sets a peak 3 dB below full scale (about 3.5 V), for normal use. **+6** sets a peak 6 dB above it (about 10 V), to drive BBD into its saturation. If the peak is too small to measure, the buttons leave the fader where it is. The peak is measured before the fader, so the result does not depend on where the fader was. If +6 moves the fader to the top, the input is too quiet to reach +6 dB; raise the level earlier in the patch.

CONNECTION

IN is the poly input, up to sixteen lanes. **L (M)** and **R** are the stereo outputs. With LAW at UNITY and every lane in the centre (VOICE SPREAD at CTR, or STACK WIDTH at its centre), the L (M) output alone carries the complete mono sum at full level.

9 · HPF

HIGH-PASS FILTER

HPF removes low frequencies from the mixed voices before the chorus. It has four fixed positions, one of which is flat, all with the same level above their cutoff. With **FREE** on, the slider moves smoothly between the positions. It runs in stereo if both the R input and the R output are patched, and in mono otherwise.



1. **Selector** four positions
2. **FREE** continuous between positions
3. **IN L** left, or mono
4. **OUT L** left
5. **IN R** right
6. **OUT R** right, or a copy of left

The four positions

The selector has four positions, from the shelf at the bottom to the strongest high-pass at the top. Each position has a curve to the right of the slider showing the filter type, and a number to the left showing its frequency or level.

Position	Label	What it is
top	**720**	first-order high-pass at 720 Hz, 6 dB per octave; makes a sound noticeably thinner
	226	first-order high-pass at 226 Hz; removes low end and leaves the rest
	FLAT	no filtering
bottom	**−2.2 dB**	low shelf: 1 dB down at 120 Hz, falling to 2.2 dB down at 20 Hz

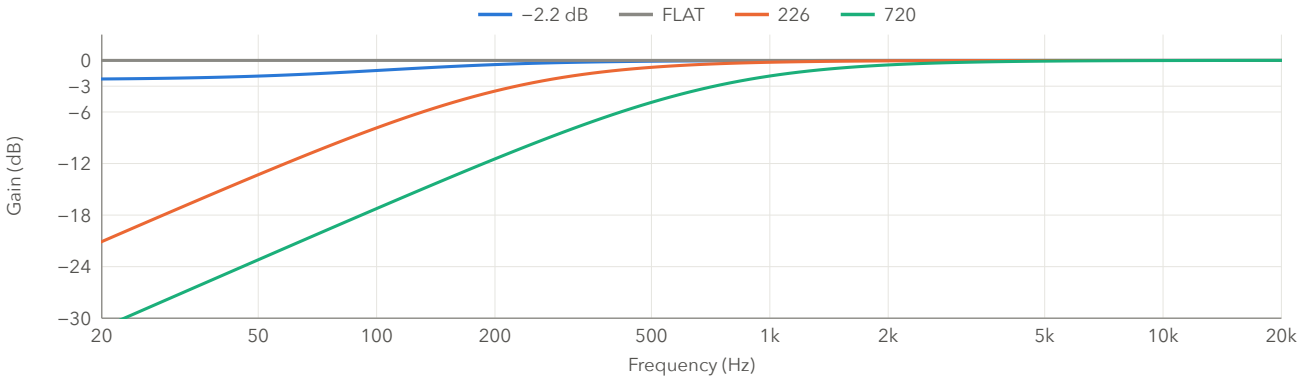


Figure 24. The four positions, measured. The high-passes are 3 dB down at 226 Hz and 722 Hz; the shelf is 2.2 dB down at the lowest frequencies. Above their cutoffs, all four are at 0 dB.

All four positions have 0 dB gain above their cutoff, so switching between them changes the tone but not the level. None of them boosts the bass: the shelf cuts the frequencies below its cutoff by 2.2 dB, rather than removing them.

The two high-passes have a gentle slope and no resonance, so they adjust the tone rather than act as a sweepable filter. 720 Hz is high enough to remove much of the body of most sounds, making them smaller and brighter, which leaves room for other sounds below.

FREE

With **FREE** on, the slider moves continuously between the positions (Figure 25).

- **Bottom third, shelf to FLAT**: the shelf's cut reduces to zero, and its corner moves up from about 120 Hz to 226 Hz.
- **Middle third, FLAT to 226**: a high-pass at 226 Hz fades in. The cutoff stays at 226 Hz; the depth increases.
- **Top third, 226 to 720**: the cutoff rises from 226 Hz to 720 Hz, at full depth. This part sounds like a filter sweep.

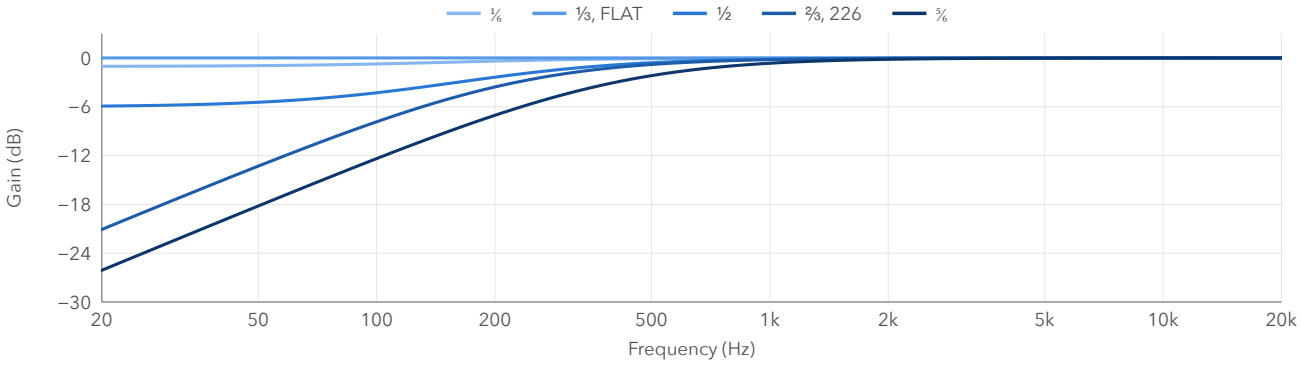


Figure 25. FREE on, with the slider at five positions, measured. At ⅙, half the shelf's cut. At ½, the 226 Hz high-pass at half depth, with a floor 6 dB down. At ⅚, the full-depth high-pass with its cutoff at 404 Hz, close to the geometric midpoint of 226 Hz and 720 Hz.

CONNECTION

HPF has four jacks: **IN** and **OUT** in two columns, **L** and **R** in two rows. There is no mono/stereo switch; HPF runs in mono or stereo depending on which jacks are patched.

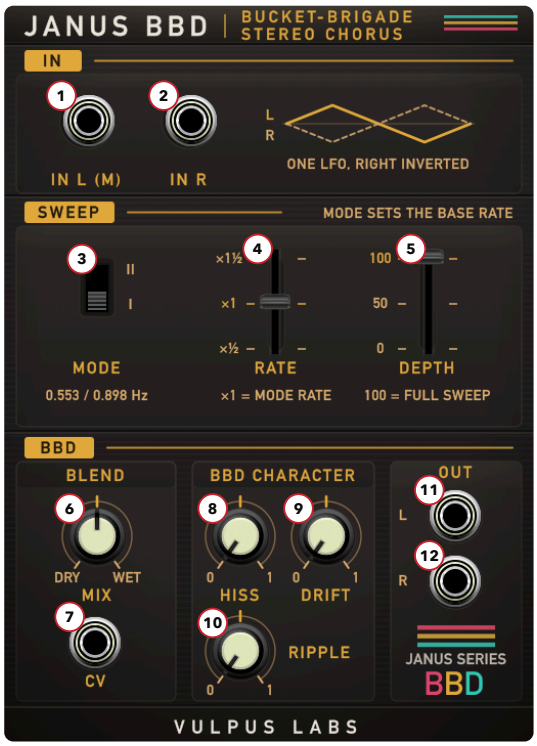
Patched	Result
L in, L out	mono
L in, L and R out	mono, with the filtered signal on both outputs
L and R in, L and R out	stereo: both channels filtered with the same settings
L and R in, L out only	mono: the right input is not used

Use L for a mono signal. In stereo, both channels always use the same settings. Using both outputs with only the left input patched copies the left output to the right, and uses no extra CPU. When you patch the right input, its channel starts from silence.

10 · BBD

BUCKET-BRIGADE STEREO CHORUS

BBD is a stereo chorus with two delay lines, whose delays are swept by one slow triangle LFO. The right line uses the LFO inverted. This makes a mono input into a wide stereo output that still sounds correct when the two outputs are mixed back to mono. Each line is a clean digital delay with the qualities of a bucket-brigade device modelled around it: its filtering, its saturation at high levels, its hiss, and two kinds of instability in its clock.



1. **IN L (M)** left, or mono
2. **IN R** right
3. **MODE** I, 0.553 Hz; II, 0.898 Hz
4. **RATE** ×½ to ×1½ the mode's rate
5. **DEPTH** width of the sweep
6. **MIX** DRY to WET
7. **CV** on MIX
8. **HISS** delay line noise
9. **DRIFT** slow random clock variation
10. **RIPPLE** 100 Hz clock variation
11. **OUT L** left
12. **OUT R** right

IN L (M) and **IN R** take a stereo pair. With only IN L patched, its signal feeds both delay lines, so a single cable is a mono input. This is the usual use. With both inputs patched, each channel goes through its own delay line.

SWEEP

MODE selects the LFO rate. **I** (down) is 0.553 Hz, a slow sweep suited to pads. **II** is 0.898 Hz, 1.62 times faster. Both modes sweep the same delay range, but II sounds deeper as well as faster, because the pitch shift of a swept delay is proportional to the sweep speed: ±7.1 cents in MODE I and ±11.5 cents in MODE II. To turn the chorus off, turn MIX to DRY.

RATE multiplies the selected mode's rate by ½ to 1½. At its centre position, the rate is the mode's own rate. The tooltip shows the multiplier.

DEPTH, 0 to 100, sets the width of the sweep around its centre. At 100, the default, the delay sweeps from 1.7 to 5.4 ms (Figure 26). Lower settings narrow the sweep around the same centre, so the tone stays the same and the movement is smaller. **0 is not bypass**: the delay stays fixed at 3.5 ms, which gives a comb-filter sound with the delay lines' filtering and saturation, but no movement.

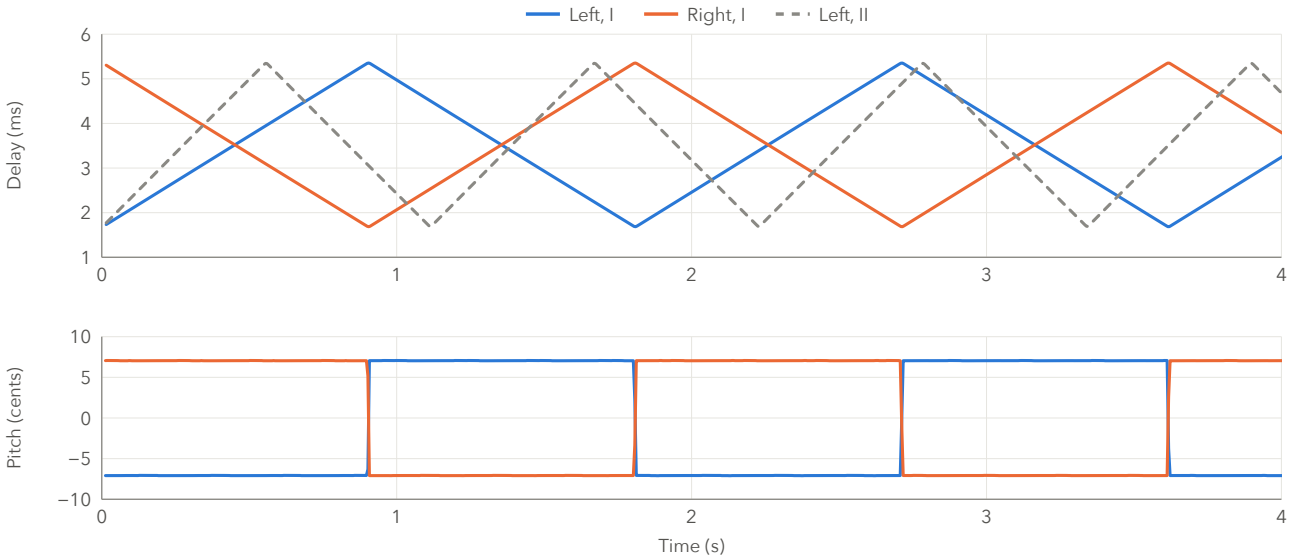


Figure 26. Above: each line's delay, measured with a test tone. The right line's delay is the left line's inverted. Below: the resulting pitch shift in MODE I, a constant 7 cents up or down that changes direction at each corner of the triangle, always opposite in the two lines.

BLEND

MIX sets the balance from **DRY** to **WET**, with an equal-power law, so the overall level stays about the same across its range. At WET the level is about 1 dB higher than at DRY. At DRY, BBD is bypassed: the input passes to the output unchanged, and the delay lines stop. Turning MIX up from DRY fades the chorus in, most of the way within 0.1 s, and turning it back to DRY fades it out the same way, so neither clicks.

CV is added to MIX. A full-scale CV covers MIX's whole range, so with MIX at DRY an envelope can fade the chorus in during a note.

BBD CHARACTER

These three knobs add imperfections of real bucket-brigade devices. **All three load at 0, where they have no effect at all.**

HISS adds the delay lines' noise. At its maximum, the noise is about 43 dB below full scale: audible in quiet passages, but well below normal signal levels. The noise level follows each line's delay: it rises as the delay gets longer and falls as it gets shorter, because a slower clock puts more of the device's noise into the audio range. The two lines' delays always move in opposite directions, so the hiss moves from side to side in stereo, but stays at a constant level in mono (Figure 27).

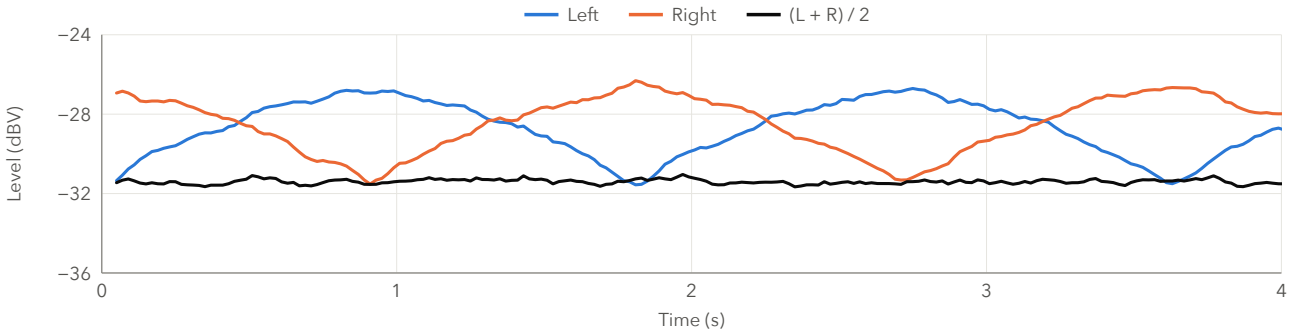


Figure 27. The hiss alone, with HISS at maximum and MODE I, measured over 100 ms windows. Each side varies by about 5 dB as its delay sweeps, in opposite directions; the mono sum varies by about 0.6 dB.

DRIFT adds slow, random variation to the delay lines' clocks. At its maximum, the pitch varies by about 2 cents on average and occasionally 5, similar to the sweep's own pitch shift but irregular. A small amount stops a long pad sounding too regular.

RIPPLE adds a periodic 100 Hz variation to the clocks, like mains hum on their power supply. It is heard as a buzz on sustained notes. At its maximum, the sidebands it creates on a 1 kHz note are 28 dB below the note. They rise with the note's pitch, about 6 dB louder for each octave up, so the buzz is stronger on high notes; a quarter of the way up is subtle. The sweep's effect and the hiss's movement cancel when the outputs are mixed to mono; the ripple does not, because it affects both lines the same way.

OUT

OUT L and **OUT R** are the stereo outputs; use both. When they are mixed to mono, the pitch modulation mostly cancels and the signal does not. This is why BBD uses one LFO inverted for the right channel, rather than two separate LFOs.

Notes on the sound

It reduces high frequencies. A bucket-brigade device filters its signal at both ends of the delay line, and BBD models this: the wet signal is flat to about 4 kHz and 3 dB down at about 6.7 kHz. This is a large part of why a bucket-brigade chorus sounds warm, where a clean digital delay sounds brighter and harsher.

Loud signals are compressed. The wet signal is 1 dB down at 3 V peak and 3 dB down at 5.7 V peak. Use a higher input level for a thicker sound, and a lower one for a cleaner sound. SUM's −3 AUTO button sets a peak level in this range; +6 aims at about 10 V peak, which is compressed by about 6 dB.

Appendix A describes how BBD works.

Starting points

Pad. MODE I, RATE at its centre, DEPTH 100, MIX two thirds of the way to WET, HISS, DRIFT and RIPPLE at 0.

Fast and wide. MODE II, DEPTH 100, RATE a little up, MIX fully WET. Suits electric pianos and plucked sounds, where the pad setting is too subtle.

Worn-out unit. DEPTH about 20, RATE down, HISS about half, DRIFT half, RIPPLE a quarter: little sweep, but plenty of noise and instability. Best on sustained single notes.

11 · Patches

Each patch starts with every control at its default. The drawing shows the cables, and the numbered controls are the ones to change, with their settings in the key below. Cables that end outside the drawing go to the rack's outputs. The first patch is a complete voice; each later patch adds a few cables to it, to vary one more control across the voices of a stack.

Several voices, and stacks of voices, add up to a higher level at SUM than one voice. After playing a few chords, press SUM's **−3** AUTO button: otherwise the outputs can go above 5 V, where Voltage clips them.

The classic voice



(1) **DCO saw**	10, as it loads	(2) **DCO sub**	4
(3) **VCF FREQ**	about half	(4) **VCF RES**	about a third
(5) **VCF ENV**	a third	(6) **VCF KYBD**	half
(7) **ENV ATTACK**	a little up	(8) **ENV SUSTAIN**	7
(9) **ENV RELEASE**	about a second	(10) **HPF**	226
(11) **BBD MIX**	two thirds wet		

KEY sends one voice per lane to DCO, and DCO's output goes through VCF. ENV controls each voice's filter cutoff and, through SUM, its level. KEY's TRIG output restarts a voice's envelope when the voice is given a new note. SUM mixes the voices to stereo, HPF reduces the low end, and BBD adds chorus. This is the voice shown in the introduction

(Figure 1), without SLU and LFO, and with KEY's V/OCT also patched to VCF's KYBD input. To have KEY prefer released voices that have finished sounding, also patch ENV's OUT + into KEY's LEVEL input.

Detuned stacks

Start from the classic voice, and set KEY's **WIDTH** to 4 and **DETUNE** to about 8 cents. Each key now plays four voices, a few cents apart, which beat against each other. Hold a note and listen: the sound swells and thins in a slow, regular sweep, the phasing effect of a detuned stack. With LAW at LIN the voices are evenly spaced, so every difference between them is a multiple of the same interval, and the beating pattern repeats exactly (Figure 28). Once in each repeat all four voices come into phase, and with them every harmonic at once: that is the strong peak in the figure, and the swell you hear. In the classic voice, DCO's sub is one octave below each saw, so the four subs are half as far apart in hertz and beat at half the rate: with the sub mixed in, the whole pattern repeats every 9.92 s.

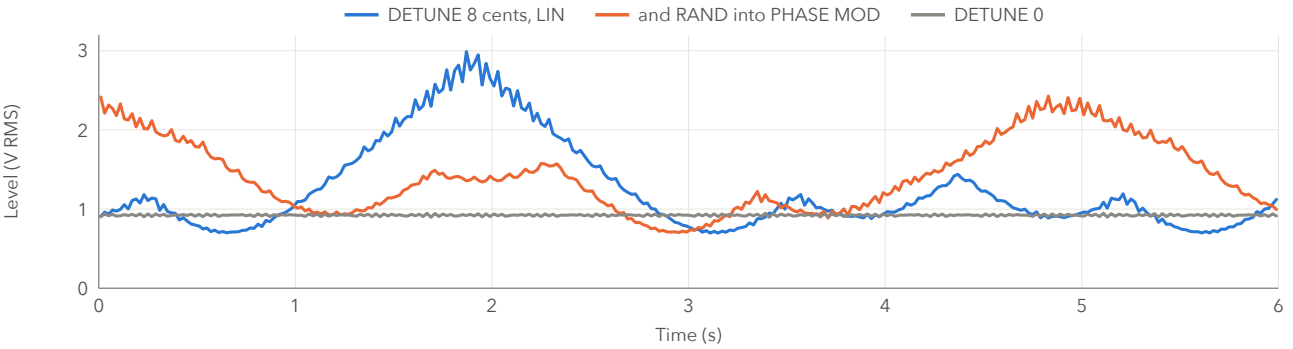


Figure 28. The level of one four-voice stack of saws at C3, measured over 20 ms windows, with the four voices averaged. With no detune, the voices have fixed phases and the level is constant. With 8 cents of detune, adjacent voices are 2.67 cents (0.2 Hz) apart, and the pattern repeats every 4.96 s: one peak when all four are in phase, and two dips, to about a quarter of the peak, where they partly cancel. With KEY's RAND fan into DCO's PHASE MOD, for one note's random phases, the pattern still repeats every 4.96 s, but the voices never come fully into phase: the highest point is 2.4 V instead of 3.0 V.

With LAW at GEOM the spacing is uneven, so the pattern takes much longer to repeat. With RAND, each note has a different pattern.

Decorrelating the stack



(1) **DCO PHASE MOD AMT** halfway right, 0.5

The voices of a stack come into phase together because they start each note in step. To stop this, patch KEY's RAND fan output into DCO's PHASE MOD **POLY CV**, and set its **AMT** halfway right. The RAND fan gives each voice a random value, held for the note, and at AMT 0.5 this moves each voice's phase by a random amount anywhere in its cycle. The detuning moves the voices' phases apart in even steps, which brings them back into line once in each repeat; random shifts are not in even steps, so after them the four voices never come fully into phase, at any harmonic. The voices still beat, so the level still rises and falls and still repeats every 4.96 s (Figure 28), but it no longer rises to a sharp peak where everything lines up: in the figure the swell is lower and broader, 2.4 V at its highest instead of 3.0 V. Each note draws new phases, so each note's pattern is different. Play the two versions one after the other to hear the difference.

Vibrato varied across a stack



1 **KEY WIDTH**	4	2 **LFO RATE**	about 5 Hz
3 **RATE AMT**	a little right, 0.14	4 **LFO DELAY**	about 2 s
5 **DELAY AMT**	a little right, 0.2	6 **LFO MODE**	IND
7 **DCO FM AMT**	a little right, 0.17		

Every voice in a stack has vibrato, but each at a slightly different rate, because the LIN fan offsets each lane's LFO rate. The four vibratos start together and slowly drift apart and back (Figure 3): with the rate offset by about ±10%, the outer two lanes of a stack at 5 Hz are half a cycle apart after about 0.5 s, and back together after about 1 s; at 1 Hz, as in the figure, it takes 2.6 s and 5.1 s. The same fan also sets a different fade-in time for each lane, so the vibrato reaches full depth on each voice at a different time (Figure 29).

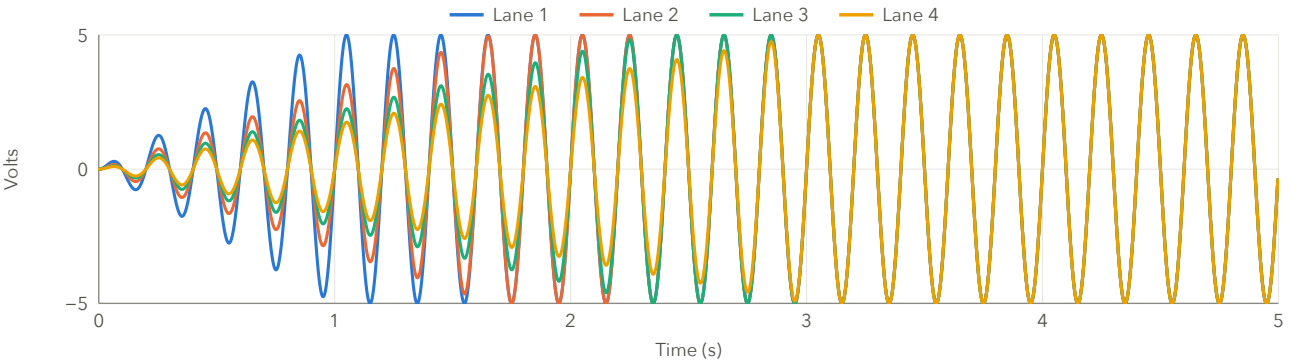
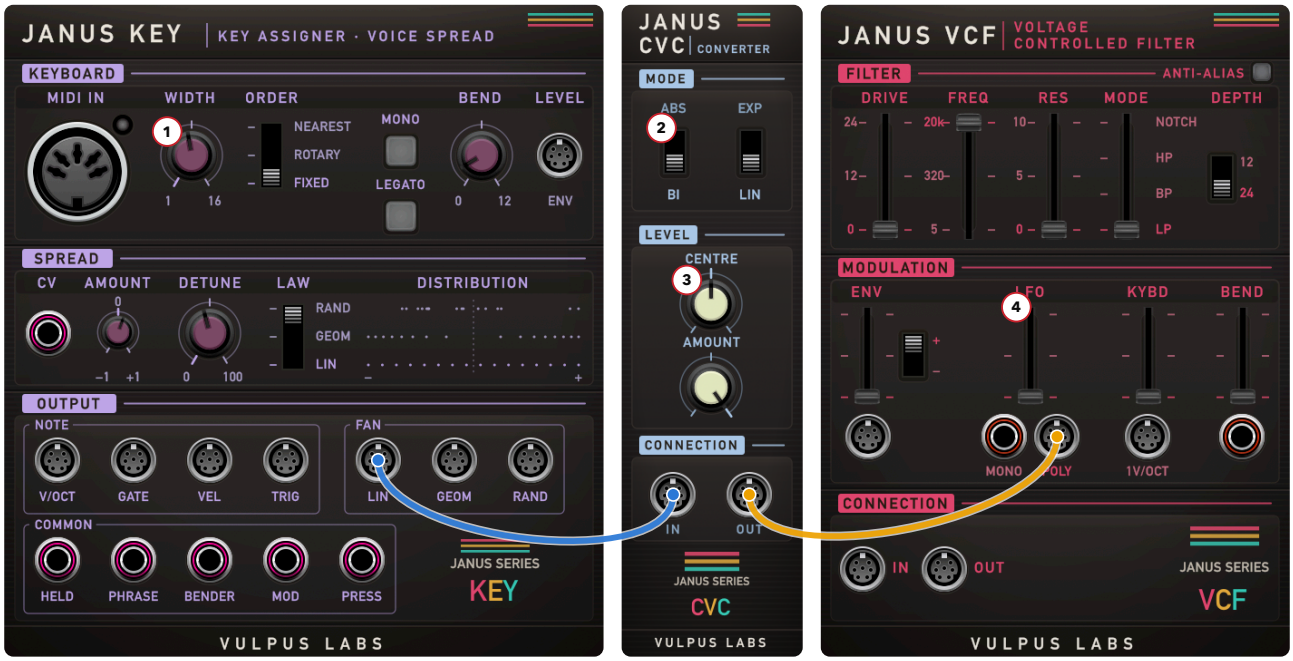


Figure 29. The four LFO lanes of one stack, measured: a 5 Hz sine, DELAY 2 s, with KEY's LIN fan into DELAY's POLY input. The lanes reach full depth after 1, 1.67, 2.33 and 3 s.

At any moment the four voices of a stack have different vibrato depths and phases, like four players holding the same note.

Brightness varied across a stack



1 **KEY WIDTH**	4	2 **CVC**	ABS
3 **CVC CENTRE**	−2.5 V	4 **VCF LFO**	about a quarter

Add this to the classic voice. CVC's ABS setting converts the fan into each voice's distance from the middle of its stack, and CENTRE at −2.5 V shifts it down, so the inner voices of each stack are darker and the outer voices brighter. This makes the stack sound wider than detuning alone. Set CVC's **AMOUNT** to −1 and CENTRE to +2.5 V for the opposite: bright inner voices and dark outer ones. With KEY's GEOM fan instead of LIN, the inner voices move closer to the outer ones, for a smaller difference between them.

Glide time varied across a stack



① **KEY WIDTH**	4	② **KEY ORDER**	NEAREST
③ **SLU RISE**	5 s	④ **Both switches**	EXP
⑤ **Both AMTs**	a little right, 0.1	⑥ **LOCK**	on

Each voice of a stack glides to a new note at a different speed, so the voices reach the new note at different times (Figure 30). ORDER at NEAREST keeps the glides short. Added to the classic voice, VCF's KYBD input still takes KEY's V/OCT, so the filter's keyboard tracking moves to the new note at once while the pitch glides; patch SLU's OUT into KYBD instead to have the filter follow the glide.

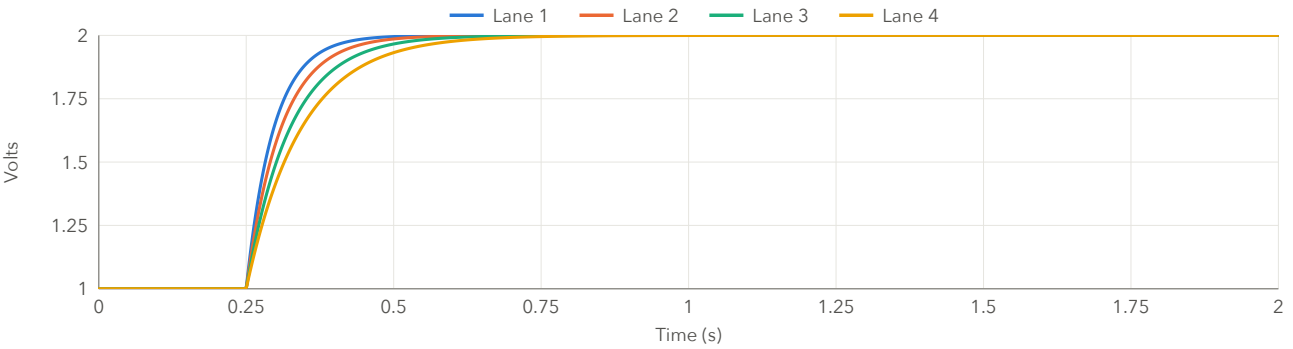


Figure 30. One stack's four pitches gliding from C3 to C4 at 0.25 s, measured, with EXP and the LIN fan into both POLY inputs. The voices arrive (within 0.5 mV) after 0.35, 0.45, 0.56 and 0.71 s; the outer two glide times differ by a factor of two.

Stereo position varied across a stack

Add to the classic voice a cable from KEY's LIN fan output into SUM's **FAN** input, and set **STACK WIDTH** fully clockwise. Each key's stack is spread from left to right, centred. With KEY's RAND fan instead, each note's voices get random positions. Combined with the patches above, each voice of a stack differs from the others in pitch, vibrato, brightness and stereo position, each by an amount you set.

A · How BBD is built

A bucket-brigade device is an analogue shift register: a chain of capacitors (the “buckets”), each passing its charge to the next at every tick of a clock. A signal written into the first bucket comes out of the last after a number of clock ticks fixed by the number of buckets, so the clock rate sets the delay: a slower clock gives a longer delay. A chorus modulates the clock rate with a slow LFO, which varies the delay. A changing delay shifts the pitch of the signal passing through it: up while the delay gets shorter, and down while it gets longer. Mixed with the dry signal, the pitch-shifted copy beats against it.

BBD does not simulate the buckets. Its delay is an ordinary digital delay line, read between samples by an interpolator that leaves the signal almost unchanged. On its own, that would be the bright, clean digital chorus that a bucket-brigade chorus does not sound like. What makes BBD sound like one is what the device does to the signal besides delaying it, and those are the parts BBD models, around the delay line:

- the low-pass filters at each end of the device, which take off the top of the signal (One line);
- the saturation where the signal is written, which compresses loud signals (One line);
- the device’s noise, which rises as the delay gets longer (The imperfections);
- the clock’s slow drift, and the ripple from its power supply (The imperfections).

The whole chorus

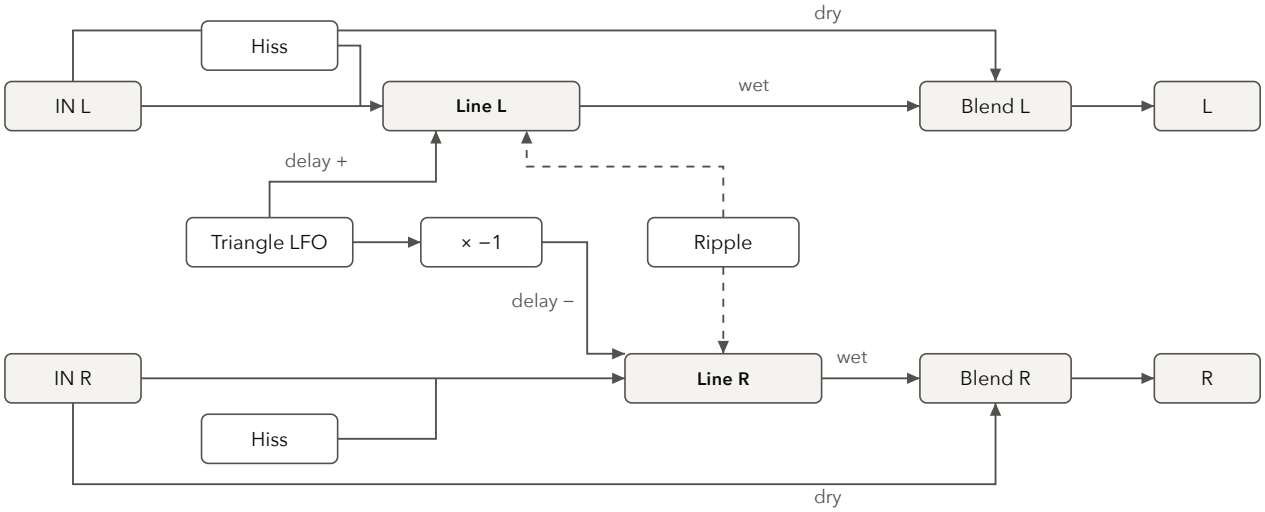


Figure 31. The chorus, with one delay line per side. One triangle LFO sets both lines’ delays, the right line’s inverted. Hiss is added at each line’s input. The ripple (dashed) changes both lines’ delays in the same direction. With only IN L patched, it feeds both lines.

The delays are

$$d_L = c + s \cdot D \cdot m(t), \quad d_R = c - s \cdot D \cdot m(t),$$

where $m(t)$ is the triangle, between -1 and $+1$; c is the centre of the sweep, 3.505 ms; s is half its width, 1.845 ms; and D is DEPTH, 0 to 1. The sweep runs from 1.66 to 5.35 ms.

One LFO, inverted, instead of two. Because the right line’s delay is the left line’s inverted, the two lines’ pitch shifts are always exactly opposite (Figure 26). When the outputs are mixed to mono, the two pitch shifts cancel, but the signal does not. Two LFOs a quarter-cycle apart would sound similar in stereo, but would not cancel in mono.

An exact triangle. The LFO’s slope is constant between its peaks and changes direction instantly at each peak, so the delay changes at a constant rate and the pitch shift is constant too: a fixed number of cents up for half a cycle, then the same number down. The pitch shift of a delay changing at rate $d'(t)$ is a factor of $1 - d'(t)$. In MODE I the delay changes by 3.69 ms in 0.9 s, which gives ±7.07 cents; in MODE II, ±11.5 cents. A sine LFO would give a pitch shift that varies continuously, spending less time at its maximum.

One line



Figure 32. One line, in the order the signal meets each stage.

Filters at both ends. A real device samples its input at the clock rate, so it needs a low-pass filter before the buckets, to prevent aliasing, and another after them, to smooth the output's steps. BBD's delay line does not sample at a clock rate, so it needs neither for that purpose; it keeps both for their effect on the sound. Both are the same filter here, built from four *poles*. A pole is one of a filter's basic responses: if the input stops, each pole's part of the output dies away at its own rate, and some also ring at their own frequency as they die. The poles are in two pairs, at $-30000 \pm 20000j$ and $-50000 \pm 30000j$ rad/s: the first number in each sets how fast it dies away, and the second how fast it rings. Both pairs die away quickly, so neither rings audibly. They are added together in parallel and scaled to pass a steady signal at exactly unity.

An analogue filter is described by equations in continuous time, and a digital filter only calculates its output once per sample, so the equations have to be converted. The usual conversions are approximations, and the error grows towards the top of the band: the filter's corner moves, by an amount that depends on the host's sample rate. Here, the input is assumed to stay constant for the length of each sample, and each pole is then moved on to the next sample by the exact solution of its equation over that interval:

$$x[n] = e^{pT} x[n-1] + \frac{e^{pT} - 1}{p} \cdot u[n],$$

where x is the pole's state, u the input, p the pole and T the length of one sample. So the poles are exactly where the analogue circuit has them, at any sample rate. The only difference from the circuit is near the top of the band, where holding the input constant for a sample is a less good description of the signal: at 48 kHz, one filter is 5.3 dB down at 20 kHz, where the circuit would be 7.7 dB down.

The poles in each pair mirror each other, so only one of each pair needs to be calculated. Because the pairs are added in parallel rather than one after the other, each pair rolls off at 6 dB per octave, like a single pole, not 12. This gentle slope is a large part of the warm sound. A 9 kHz one-pole low-pass after the output filter reduces the highest frequencies further. Through a whole line, the response is a fraction of a decibel up near 2.4 kHz and 3 dB down at 6.7 kHz (Figure 33).

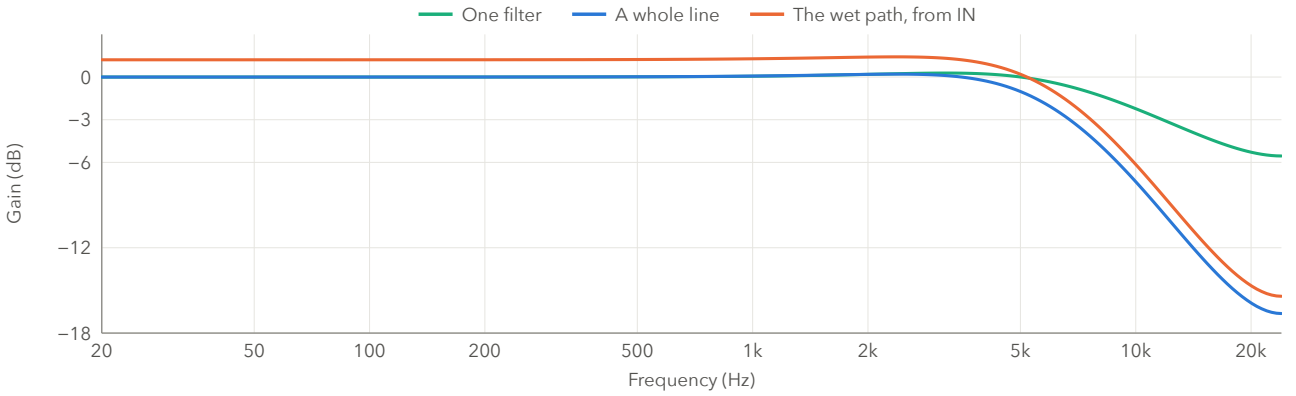


Figure 33. One filter alone, computed from its poles at 48 kHz: 3 dB down at 11.8 kHz and 5.3 dB at 20 kHz. A whole line, measured with both filters and the trim: 3 dB down at 6.7 kHz, 15.9 dB at 20 kHz. The module's wet path at MIX WET sits 1.2 dB higher, the wet leg's gain.

Saturation at the input. A bucket can only hold a limited charge, so the signal written into the buckets passes through a soft saturation curve, $\tanh(1.2x)/1.2$, which has a gain of 1 for small signals and compresses large ones. As in a real device, the saturation is applied after the input filter.

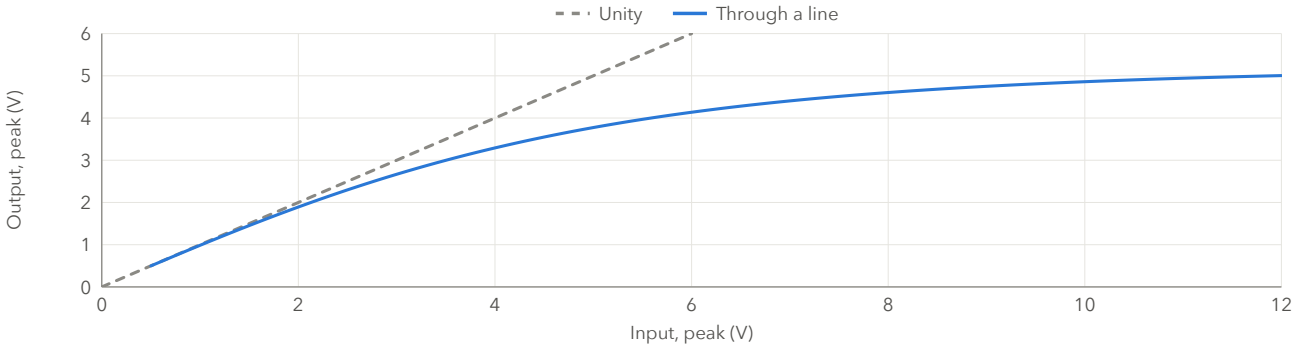


Figure 34. The fundamental of a 50 Hz sine through the wet path, DEPTH 0, the wet leg's gain divided out. 1 dB down at 3 V peak, 3 dB at 5.7 V, and at 12 V, 7.6 dB.

Fractional delay. This is where the buckets would be, and it is meant to change the signal as little as possible: the colour comes from the stages around it. The required delay is rarely a whole number of samples, and it changes every sample. The output is read with a first-order all-pass interpolator,

$$y[n] = a(x_N - y[n-1]) + x_{N+1}, \quad a = \frac{1-f}{1+f},$$

for a delay of $N + f$ samples. Its gain is exactly 1 at every frequency, and its delay is close to f at low and middle frequencies, rising towards the top of the band: at half a sample it is 0.54 samples at 5 kHz and 0.69 at 10 kHz. Linear interpolation between two samples gives the same delay, but reduces high frequencies: at half a sample it is 6 dB down at 16 kHz and silent at the top of the band (Figure 35). As the sweep moves the fraction between 0 and 1, this loss would come and go. The fraction is kept at least a thousandth of a sample from 0 and 1, because at 0 the all-pass coefficient reaches 1 and the filter becomes unstable.

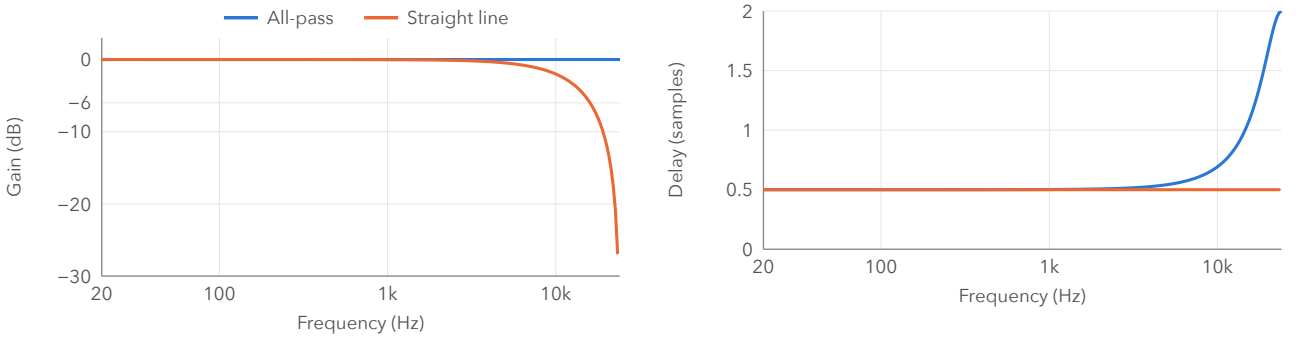


Figure 35. Two ways to read a delay of half a sample. The all-pass keeps every frequency's level, and delays low and middle frequencies by half a sample, and high frequencies by more. Linear interpolation keeps the delay but loses high frequencies.

The imperfections

Hiss. Noise is added at each line's input, before the filters, so it is filtered like the signal. Its level depends on the line's delay: a device makes the same noise at every transfer, and a slower clock concentrates it in a narrower frequency range, so more of it is in the audio range. The noise power is proportional to the delay, so its amplitude is scaled by

$$\sqrt{d/c},$$

which is 0.69 at the shortest delay and 1.24 at the longest, a range of 5 dB, and the noise power it gives averages exactly 1 over a sweep. The two lines are always at opposite ends of the sweep, so the hiss moves from side to side in stereo and stays constant in mono (Figure 27).

Drift. A clock built from a few transistors changes slowly and randomly with temperature and supply voltage. Each line's delay is multiplied by $1 + 0.1 \cdot \text{DRIFT} \cdot w(t)$, where w is a random walk between -1 and $+1$ that takes a small

step every 64 samples. Between steps, w is interpolated linearly. Without interpolation the delay would change in steps, and a delay that is not changing does not shift the pitch, so the drift would be inaudible. The two lines' random walks are independent (Figure 36).

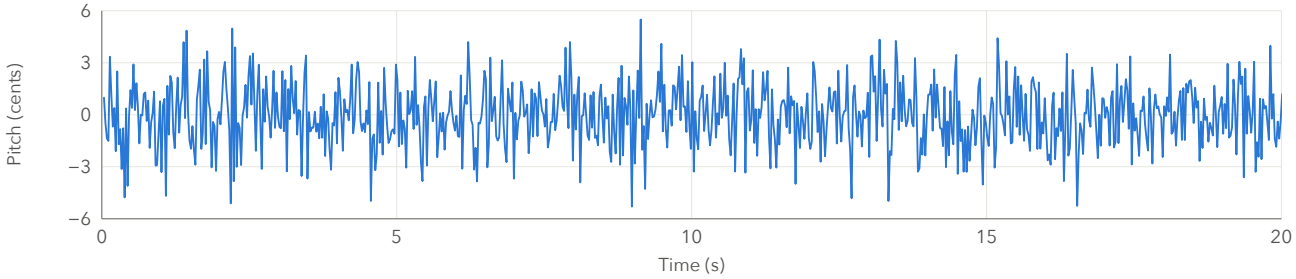


Figure 36. The left line's pitch with DRIFT at the top and DEPTH at 0, averaged over 25 ms: 1.8 cents RMS, peaking at 5.5.

Ripple. A power supply rectified from 50 Hz mains drops slightly between the mains' peaks and is recharged at each peak, twice per cycle, so the clocks vary periodically at 100 Hz. This is modelled as an asymmetric triangle wave, falling for 85% of each cycle and rising in the remaining 15%. The asymmetry adds harmonics of 100 Hz, which makes the effect sound like a buzz rather than a vibrato. It multiplies both lines' delays by the same $1 + 0.005 \cdot \mathrm{RIPPLE} \cdot r(t)$, because both clocks share one supply. This varies the delay, so it is a frequency modulation, and the sidebands it makes rise with the note's frequency. At maximum, on a 1 kHz note, the first sidebands are 28 dB below the note (Figure 37).

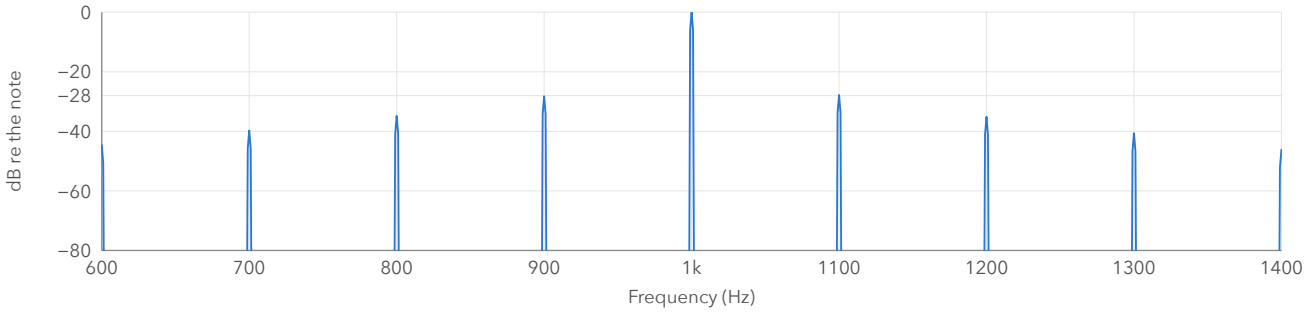


Figure 37. A 1 kHz tone through the wet path, DEPTH 0, RIPPLE at the top: sidebands every 100 Hz, the first pair at −28 dB, the second at −35, the third at −40.

The blend

The dry and wet signals are mixed at equal power: the dry gain is $\sqrt{1-w}$ and the wet gain $1.15\sqrt{w}$, where w is the wet proportion from MIX and its CV. The factor 1.15 makes the wet signal slightly louder. The square roots are calculated every 64 samples, and the gains are interpolated linearly in between, so moving MIX does not cause steps. w itself follows MIX through a one-pole smoother with a 30 ms time constant, so switching the chorus in or out does not click. When w reaches 0 (DRY), the delay lines stop and the input passes straight to the output.

B · How DCO is built

An oscillator timed digitally does not drift, because its frequency is set by a count, not by a voltage on a capacitor. DCO is digitally timed in the same way, and generates its three waveforms from one ramp, as such an oscillator does, rather than calculating each separately.

One ramp per voice

Each voice's phase is a 32-bit whole number, a fraction of a cycle: 0 is the start of the cycle and 2^{32} would be its end. Every sample the phase is advanced by a step proportional to the frequency, and when it passes 2^{32} it overflows back to near zero. So the cycle restarts automatically, with no extra code, and the phase can never go out of range. The step is worked out every 32 samples, 1.5 kHz at 48 kHz, and ramped between, so that pitch modulation is smooth. It is held under half a cycle per sample.

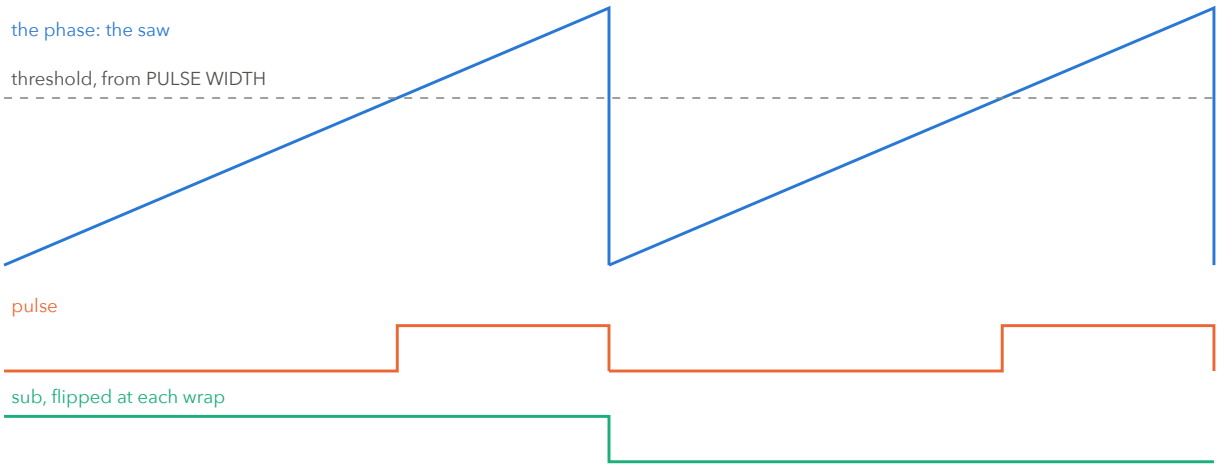


Figure 38. Three waveforms from one phase. The saw is the phase itself. The pulse is high while the phase is above a threshold, so it rises part-way up the ramp and falls with the wrap. The sub changes state at every wrap: a square at half the frequency, always in step with the saw.

- **The saw is the phase.** With its top bit inverted and read as a signed number, the phase runs from its most negative value to its most positive over each cycle, which is an exact ramp.
- **The pulse is one comparison** of the phase against a threshold, $1 - w$ of the way up for a width w. The width is kept between 5% and 95%, and further from the wrap at high pitches, up to 25% to 75% above 6 kHz, so that the pulse's two edges are always at least two samples apart.
- **The sub is a flip-flop** that changes state at each wrap, so it is an exact square at exactly half the frequency.

Because the pulse falls at the saw's wrap, saw and pulse together make a stepped saw, and WIDTH moves the step along the ramp (Figure 7). With ANTI-ALIAS off, above a quarter of a cycle per sample (12 kHz at 48 kHz), all three waveforms change gradually into sine waves, completely so by 21.6 kHz, because no higher harmonics fit below the top of the band. With ANTI-ALIAS on, the filter described below removes those harmonics instead.

Smoothing the edges

A ramp's reset and a pulse's edges are instant jumps. A sampled jump contains harmonics far above half the sample rate, which are reflected back into the audio range as inharmonic tones: aliasing. DCO corrects each jump by adding a small correction to the samples on either side of it, so that the samples near the edge match a band-limited waveform. The correction is a polynomial in the jump's exact position between samples, which is known from the phase. This is the PolyBLEP method.

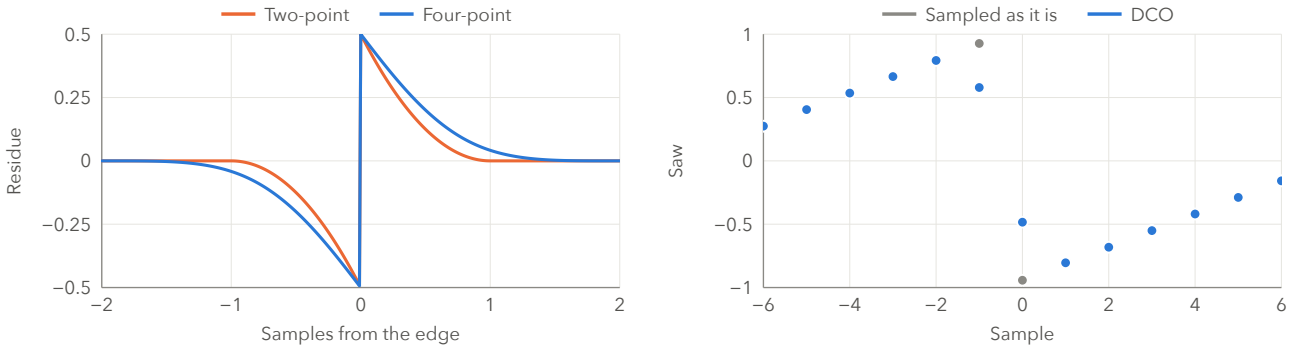


Figure 39. Left: the correction added around a downward jump, for the two-point method and the four-point method DCO uses. Right: a saw at 3.1 kHz around one reset, which happens 0.44 of a sample before sample 0, measured. Only the four samples nearest the jump are changed.

The four-point correction is an integrated cubic B-spline, covering two samples on each side of the edge. Its spectrum falls off much faster than the two-point correction's. On a saw at A6, the loudest alias below 15 kHz is −25.7 dB with no correction, −42.9 dB with two-point correction and −60.0 dB with four-point, measured. The cost is a slight loss of level in the top octave, which matters little when the oscillator is followed by a low-pass filter.

Through-zero FM runs the phase backwards, with a negative step. The same code serves, with the edges' directions swapped.

Phase modulation adds an offset to the phase used to calculate the output, not to the stored phase. The corrections take the changing offset into account, so the edges stay clean under audio-rate phase modulation. The step is a 32-bit number, so it can never exceed half a cycle per sample: deeper modulation wraps round to a smaller step in the other direction, and the correction still applies.

Sync

The exact time of a rising zero crossing at SYNC is found by linear interpolation between two samples, and the voice's phase is reset at that time. The correction needs the sample before the jump as well as the one after, but an external signal cannot be predicted, so the reset is applied one sample (21 μs) late. The sub is reset to a fixed state at the same time. Sync's jump uses the two-point correction, because only that fits within one sample.

Voices, noise, and the output

Different start phases. Voice v starts at $v \times 0.618\ldots$ of a cycle (the fractional part of the golden ratio), which spreads the voices' start phases around the cycle. So two voices on the same note do not simply add up to a louder copy of one (Figure 10), and the start phases are the same every time.

One noise source. A single generator makes one noise sample per host sample, shared by all sixteen voices. So N voices give N times the level of one, not $\sqrt{N}$ times.

A coupling capacitor. Each voice's mix passes through a one-pole high-pass filter at 5 Hz, which removes the pulse's DC offset. The offset changes with the pulse width, and when the width changes quickly, some of the change passes through before the filter removes it, causing a slight thump. At low pitches the same filter visibly curves the saw and the sub (Figure 7).

ANTI-ALIAS

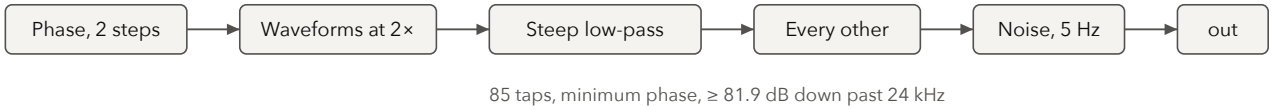


Figure 40. The double-rate path. Each host sample's phase step is split into two halves that add up to it exactly, so the phase is the same as the normal path's at every host sample.

With ANTI-ALIAS on, each voice calculates two samples per host sample, advancing its phase by half a step each time, with phase and pulse-width modulation interpolated between them. A steep minimum-phase low-pass filter (85 taps; within 0.19 dB up to 19 kHz, −6 dB at 21 kHz, and at least 81.9 dB down above 24 kHz) then removes everything that would alias, and every other sample is kept. The noise and the coupling capacitor are applied after this, at the host's rate. The filter delays the output by 1.5 samples over most of the audio range. At A4, aliasing below 15 kHz falls from −67.9 dB to −90.1 dB.

When the button is pressed, the newly selected path starts running a little early, without being heard, so that its filter is filled with signal; then the output crossfades between the two paths over 20 ms.